Open Problems in List Decoding and Correlated Agreement

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Abstract

The Ethereum Foundation recently announced *the Proximity Prize*[1] which aims to resolve some open problems that play an important role in the design of succinct proof systems. This paper reviews the open problems relevant to the Proximity Prize. We focus on some grand challenges relating to list decoding bounds, proximity gaps, correlated agreement, and mutual correlated agreement, as they relate to proof systems and Reed–Solomon codes. Along the way we survey the known results on these topics.

Keywords: proximity gaps, correlated agreement, list decoding, proximity prize

[1] https://proximityprize.org/

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1 Introduction

Recent progress in succinct non-interactive proof systems, called SNARKs, gives rise to new concepts in coding theory called *proximity gaps* [RVW13; AHIV17], *correlated agreement* (CA) [BGKS20; BCIKS20], and *mutual correlated agreement* (MCA) [ACFY25]. These SNARK systems also rely on classic *list decoding* bounds from coding theory [Eli57; Woz58]. A better understanding of these concepts will lead to more efficient SNARKs, and this gives rise to a number of fascinating open problems that are both mathematically interesting and important in practice.

The goal of this paper is to explicitly list these open problems and to survey our present state of knowledge. This is a fast-moving field with a rapid stream of new results. This paper covers the known results at the time of this writing.

List decoding bounds. First, recall that a ***linear code*** $\mathcal{C}$ is a linear subspace of $\mathbb{F}^n$, where $\mathbb{F}$ is a finite field, and n is called the code length. For a word $f \in \mathbb{F}^n$ and $\delta \in [0,1]$ we write $\Lambda(\mathcal{C}, \delta, f)$ for the set of codewords in $\mathcal{C}$ whose Hamming distance from f is at most δn. The list decoding bound of $\mathcal{C}$ is defined as

$$\Lambda(\mathcal{C}, \delta) \coloneqq \max_{f \in \mathbb{F}^n} \left|\Lambda(\mathcal{C}, \delta, f)\right|.$$

The main list decoding question of relevance to this paper is to find tight upper bounds on $\Lambda(\mathcal{C}, \delta)$ for an interleaved Reed–Solomon code (as defined in Section 2.3). In particular, the goal is to understand for what values of δ the list $\Lambda(\mathcal{C}, \delta)$ is "small." In Section 3 we survey the known results and open questions on this topic that are most relevant to the development of succinct proof systems.

Proximity gaps. We next explain the notions of a proximity gap, correlated agreement (CA), and mutual correlated agreement (MCA). We begin with some useful terminology.

- For $\delta \in [0,1]$ we say that $g \in \mathbb{F}^n$ is ***δ-close to $\mathcal{C}$*** if there is a codeword $c \in \mathcal{C}$ whose Hamming distance from g is at most δn. Furthermore, for $S \subseteq [n]$ we say that $g \in \mathbb{F}^n$ is ***(S, δ)-close to $\mathcal{C}$*** if $|S| \geq (1-\delta)n$ and there is a codeword $c \in \mathcal{C}$ such that g and c agree on S, namely $g_i = c_i$ for all $i \in S$. Clearly, if g is (S, δ)-close to $\mathcal{C}$ then g is δ-close to $\mathcal{C}$. Conversely, if g is δ-close to $\mathcal{C}$ then there is a set $S \subseteq [n]$ such that g is (S, δ)-close to $\mathcal{C}$.

- For a set $L \subseteq \mathbb{F}^n$ we say that L is δ-close to $\mathcal{C}$ if every element in L is δ-close to $\mathcal{C}$. Similarly we say that L is (S, δ)-close to $\mathcal{C}$ if every element of L is (S, δ)-close to $\mathcal{C}$. Observe that if $g_1, \ldots, g_\ell \in \mathbb{F}^n$ are all (S, δ)-close to $\mathcal{C}$ then the entire linear span of $g_1, \ldots, g_\ell$ in $\mathbb{F}^n$ is (S, δ)-close to $\mathcal{C}$.

- We define proximity gaps, correlated agreement, and mutual correlated agreement with respect to a family of sets $\mathcal{F} \subseteq 2^{(\mathbb{F}^n)}$. Some set families of interest include
  - Lines, namely $\mathcal{F}_{\text{lines}} \coloneqq \left\{ \{f_1 + \gamma f_2\}_{\gamma \in \mathbb{F}} \;\middle|\; f_1, f_2 \in \mathbb{F}^n \right\}$.
  - Affine spaces, namely $\mathcal{F}_{\text{spaces}} \coloneqq \left\{ \{f_0 + \gamma_1 f_1 + \ldots \gamma_\ell f_\ell\}_{\gamma_1, \ldots, \gamma_\ell \in \mathbb{F}} \;\middle|\; f_0, \ldots, f_\ell \in \mathbb{F}^n \right\}$.
  - Power curves, namely $\mathcal{F}_{\text{curves}} \coloneqq \left\{ \{f_0 + \gamma f_1 + \ldots \gamma^\ell f_\ell\}_{\gamma \in \mathbb{F}} \;\middle|\; f_0, \ldots, f_\ell \in \mathbb{F}^n \right\}$.

  Throughout the paper we will primarily use the family $\mathcal{F}_{\text{lines}}$, but all the open problems we discuss can be similarly stated for other families of sets [2]. It is not difficult to see that a line $L = \{f_1 + \gamma f_2\}_{\gamma \in \mathbb{F}}$ is (S, δ)-close to $\mathcal{C}$ if and only if both f_1 and f_2 are (S, δ)-close to $\mathcal{C}$.

[2] See [BCGM25] for a discussion in the setting of polynomials generators.

With this terminology in place, we can now explain the three concepts: proximity gap, correlated agreement, and mutual correlated agreement for a code $\mathcal{C} \subseteq \mathbb{F}^n$ with respect to a family $\mathcal{F} \subseteq 2^{(\mathbb{F}^n)}$.

- **Proximity gap:** For $\delta \in [0,1]$ and $L \in \mathcal{F}$, let $p(L) \in [0,1]$ be the fraction of points $g \in L$ such that g is δ-close to $\mathcal{C}$. We say that $\mathcal{C}$ has ***proximity gap*** error $\varepsilon_{\mathsf{pg}}(\mathcal{C},\delta) \in [0,1]$ if for every $L \in \mathcal{F}$ such that $p(L) > \varepsilon_{\mathsf{pg}}(\mathcal{C},\delta)$, we have that L is δ-close to $\mathcal{C}$ (which implies that $p(L) = 1$). In other words, there is a dichotomic gap: either $p(L) \leq \varepsilon_{\mathsf{pg}}(\mathcal{C},\delta)$ or $p(L) = 1$.

- **Correlated agreement (CA):** CA provides a stronger guarantee than a proximity gap: when $p(L) > \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$, not only is L δ-close to $\mathcal{C}$, but there is an $S \subseteq [n]$ such that L is (S,δ)-close to $\mathcal{C}$. More precisely, we say that $\mathcal{C}$ has ***correlated agreement*** error $\varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) \in [0,1]$ if for every $L \in \mathcal{F}$ such that $p(L) > \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$, we have that L is (S,δ)-close to $\mathcal{C}$ for some $S \subseteq [n]$.

- **Mutual correlated agreement (MCA):** MCA provides a stronger guarantee than CA: when $p(L) > \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$, not only is L (S,δ)-close to $\mathcal{C}$ for some S, it has to be (S,δ)-close for the same S's as the individual points on L. More precisely, for $\delta \in [0,1]$ and $L \in \mathcal{F}$, we say that a point $g \in L$ is "good" if for every S such that g is (S,δ)-close to $\mathcal{C}$ it holds that L is (S,δ)-close to $\mathcal{C}$. We say that $\mathcal{C}$ has ***mutual correlated agreement*** error $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$ if for every $L \in \mathcal{F}$ at most $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$ of the points on L are bad.

We give the complete definitions in Section 4.1 where we also allow for a proximity loss between the pre-condition and the conclusion. Because each notion is a strengthening of the previous, it should be clear that for every linear code $\mathcal{C}$ and distance δ we have

$$\varepsilon_{\mathsf{pg}}(\mathcal{C},\delta) \leq \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) \leq \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta).$$

What is MCA good for? MCA is an important property in the design of efficient SNARK systems. In Section 6 we present an illustrative toy protocol whose knowledge soundness error is directly related to the MCA error and list decoding bound of some underlying code $\mathcal{C}$. Briefly, suppose the prover is committed to two words $f_1, f_2 \in \mathbb{F}^n$. It wants to convince the verifier that f_1, f_2 are close to codewords that encode messages $\boldsymbol{f}_1, \boldsymbol{f}_2 \in \mathbb{F}^k$, and that both these messages satisfy some public linear constraint. This problem is a key component in some proof systems such as WHIR [ACFY25] and others. The MCA property of $\mathcal{C}$ lets us argue that a simple and natural protocol for this problem, described in Section 6, is knowledge sound. The weaker correlated agreement (CA) property lets us argue that this natural protocol proves that f_1, f_2 are close to some codewords in $\mathcal{C}$, but as far as we know, it does not let us prove properties of the messages encoded by these close-by codewords. For that we need MCA, although one can alternatively use a stronger notion called *line-decoding* (Section 4.4).

The grand MCA challenge. From here we will focus almost exclusively on the Reed–Solomon code $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F},\mathcal{L},k]$ where $\mathcal{L}$ is a multiplicative subgroup of $\mathbb{F}$ whose size is a power of two, since that is a widely used code in SNARK systems.[3] Moreover, for simplicity we will only consider MCA with respect to the family of lines $\mathcal{F}_{\text{lines}}$.

The design of SNARK systems requires a detailed understanding of how the MCA error $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$ of the Reed–Solomon code $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F},\mathcal{L},k]$ behaves as δ varies in $[0,1]$. Table 1 summarizes what is known, and Section 4.2 goes into far more detail. Here $n \coloneqq |\mathcal{L}|$ is the code length, $\delta_{\mathsf{min}}(\mathcal{C}) \in [0,1]$

[3] Recall that the Reed–Solomon code $\mathsf{RS}[\mathbb{F},\mathcal{L},k]$ is the set of all functions $f\colon \mathcal{L} \to \mathbb{F}$ where there is a degree $< k$ univariate polynomial $\hat{f} \in \mathbb{F}^{<k}[X]$ with $f(x) = \hat{f}(x)$ for every $x \in \mathcal{L}$.

denotes the relative minimum distance of $\mathcal{C}$, and $J(\delta_{\mathsf{min}}(\mathcal{C})) := 1-\sqrt{1-\delta_{\mathsf{min}}(\mathcal{C})}$ is the Johnson radius of $\mathcal{C}$. Note that for all codes, $J(\delta_{\mathsf{min}}(\mathcal{C})) \leq \delta_{\mathsf{min}}(\mathcal{C})$. We will refer to $\delta_{\mathsf{min}}(\mathcal{C})$ as the capacity of $\mathcal{C}$.

The grand challenge is to resolve how $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$ behaves when δ is between the Johnson radius of $\mathcal{C}$ and the capacity of $\mathcal{C}$. We state this very concretely as the following open question. The challenge refers to a *smooth evaluation domain* $\mathcal{L}$, which is a multiplicative subgroup of $\mathbb{F}$ whose size is a power of two (Definition 2.12).

The grand MCA challenge. We are given a Reed–Solomon code $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ defined over some smooth evaluation domain $\mathcal{L} \subseteq \mathbb{F}$. The code has constant rate, and in particular the rate $\rho(\mathcal{C}) := k/|\mathcal{L}|$ is one of $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$. For a given ϵ^*, say $\epsilon^* = 2^{-128}$,

$$\text{determine the largest } \delta^*_{\mathcal{C}} \in [0,1] \text{ such that } \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta^*_{\mathcal{C}}) \leq \epsilon^*,$$

assuming $|\mathbb{F}|$ is sufficiently large so that such a $\delta^*_{\mathcal{C}}$ exists.

Resolving the challenge for a code $\mathcal{C}$ and ϵ^* requires specifying a $\delta^*_{\mathcal{C}} \in [0,1]$ along with a proof that for all $\delta > \delta^*_{\mathcal{C}}$ we have $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) > \epsilon^*$. Ideally we could determine this $\delta^*_{\mathcal{C}}$ for every Reed–Solomon code $\mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$, namely, for every choice of $\mathbb{F}$, $\mathcal{L}$, and k. However, we are mostly interested in determining $\delta^*_{\mathcal{C}}$ when $\mathcal{L} \subseteq \mathbb{F}$ is a smooth domain, $k \leq 2^{40}$, and $|\mathbb{F}| < 2^{256}$.

Roughly speaking, the results in Table 1 show that when $|\mathcal{L}|/|\mathbb{F}| < \epsilon^*$ then $\delta^*_{\mathcal{C}}$ is at least the Johnson radius $J(\delta_{\mathsf{min}}(\mathcal{C}))$ minus some small constant. The challenge is to determine when is $\delta^*_{\mathcal{C}}$ greater than the Johnson radius. In particular, for which Reed–Solomon codes of interest, is $\delta^*_{\mathcal{C}}$ close to the capacity radius of the code? One way to approach this challenge is using the concept of line-decoding and its connection to MCA discussed in Theorem 4.21.

The grand list decoding challenge. In addition to bounds on MCA for a code $\mathcal{C}$, one often also requires a good list decoding bound for $\mathcal{C}$. We will see this explicitly in Section 6. We therefore state the following list decoding grand challenge, as needed for protocol analysis. The challenge is stated as it applies to an m-way interleaved Reed–Solomon code (Definition 2.9) denoted by $\mathcal{C}^{\equiv m}$.

The grand list decoding challenge. We are given a Reed–Solomon code $\mathcal{C}$ as in the grand MCA challenge. For a given ϵ^*, say $\epsilon^* = 2^{-128}$, and a constant m,

$$\text{determine the largest } \delta^*_{\mathcal{C}} \in [0,1] \text{ such that } |\Lambda(\mathcal{C}^{\equiv m}, \delta^*_{\mathcal{C}})| \leq \epsilon^* \cdot |\mathbb{F}|,$$

assuming $|\mathbb{F}|$ is sufficiently large so that such a $\delta^*_{\mathcal{C}}$ exists.

The reason for the bound $\epsilon^* \cdot |\mathbb{F}|$ on the maximum list size is that the soundness error of coding based proof systems often includes a term of the form $|\Lambda(\mathcal{C}^{\equiv m}, \delta)|/|\mathbb{F}|$, as illustrated in Section 6. Hence, we are interested in the largest value of δ for which this expression is negligible, or concretely less than $\epsilon^* = 2^{-128}$.

Resolving the challenge for a code $\mathcal{C}$, an m, and an ϵ^*, requires specifying a $\delta^*_{\mathcal{C}} \in [0,1]$ along with a proof that for all $\delta > \delta^*_{\mathcal{C}}$ we have $|\Lambda(\mathcal{C}^{\equiv m}, \delta^*_{\mathcal{C}})| > \epsilon^* \cdot |\mathbb{F}|$. Again, the key question is whether for the Reed–Solomon codes of interest, $\delta^*_{\mathcal{C}}$ is close to the capacity radius of the code. We stress that we do not need an efficient list decoding algorithm; we are only interested in the value of $\delta^*_{\mathcal{C}}$.

What is in the rest of the paper. After some preliminaries in Section 2, we present in Section 3 what is known and what is open about the list decoding grand challenge. In Section 4 we present

distance δ	error $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$	reference
$\delta = 0$	$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) = \frac{2}{\lvert\mathbb{F}\rvert}$	trivial
$\delta < \delta_{\mathsf{min}}(\mathcal{C})/2$ (δ below the unique decoding radius)	$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \le \frac{O(n)}{\lvert\mathbb{F}\rvert}$	[ACFY25; BCIKS20]
$\delta = J(\delta_{\mathsf{min}}(\mathcal{C})) - \eta$ (as δ approaches the Johnson radius)	$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \le \frac{n \cdot \mathrm{poly}(1/\eta)}{\lvert\mathbb{F}\rvert}$	[BCHKS25; Hab25; BCGM25; BCIKS20]
$\delta \approx \delta_{\mathsf{min}}(\mathcal{C}) - \frac{1}{\Omega(\log n)}$ (as δ approaches capacity)	$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \ge n^{\Omega(1)}/\lvert\mathbb{F}\rvert$	[BCHKS25; KK25; CGHLL26], for a large enough $\mathbb{F}$

Table 1: The behavior of $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$ for $\delta \in [0, 1]$ and $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$, where $n \coloneqq \lvert\mathcal{L}\rvert$.

what is known and what is open about the MCA grand challenge. In Section 5 we explore the known connections between list decoding and correlated agreement. One might hope that in the cases of interest to us, a good list decoding bound implies a good MCA bound, so that both our grand challenges collapse to one. While this is currently an open problem, some connections between these two notions are known and presented in Section 5. In Section 6 we describe a toy problem and a toy protocol that explain why our two grand challenges are useful for building a succinct proof system. Finally, in Section 7 we discuss a number of interesting open problems related to our grand challenges that would also be good to solve.

2 Preliminaries

We use the following convention and notation:

- Logarithms are assumed to be base 2 unless otherwise specified.
- The "hat" symbol over a function (e.g. $\hat{p}$) denotes a polynomial.
- We interchangeably interpret vectors $\boldsymbol{f} \in \Sigma^n$ as functions $f\colon [n] \to \Sigma$.
- For a field $\mathbb{F}$, a variable count $m \in \mathbb{N}$, and an individual degree bound $d \in \mathbb{N}$, we let $\mathbb{F}^{<d}[X_1, \dots, X_m]$ be the set of all m-variate polynomials over $\mathbb{F}$ of individual degree at most $d - 1$ with variables $(X_1, \dots, X_m)$.
- In a random process, for a set S, we let $x \leftarrow S$ represent that x is a random variable whose distribution is uniform over S (we think of it as "sampling x uniformly from the set S"). Similarly, if A is a randomized algorithm, we use $x \leftarrow A$ to denote that x is a random variable distributed as the output distribution of A.

We recall the polynomial identity lemma:

Lemma 2.1 (Polynomial identity lemma)**.** *Let $\hat{p} \in \mathbb{F}^{<d}[X_1, \dots, X_m]$ be a non-zero polynomial. Then,*

$$\Pr_{\boldsymbol{v} \leftarrow \mathbb{F}^m}[\hat{p}(\boldsymbol{v}) = 0] \le \frac{m \cdot (d-1)}{\lvert\mathbb{F}\rvert} .$$

2.1 Probability

We introduce notation for defining random variables and events over these, following [CY24].

- We write $\{x \mid x \leftarrow A\}$ to denote the random variable that consists of the string x sampled according to the probabilistic algorithm A.
- We write $\Pr[f(x) = 1 \mid x \leftarrow A]$ to denote the probability of the event that $f(x) = 1$ when x is sampled according to the probabilistic algorithm A, for a given boolean function f.
- We write $\Pr[D(x) = 1 \mid x \leftarrow A]$ to denote the probability of the event that $D(x) = 1$ when x is sampled according to the probabilistic algorithm A, for a given probabilistic algorithm D. In this case the probability is over the randomness of both A and D.

Sometimes we implicitly define the algorithm A or the boolean function *within the expression itself*, by using pseudocode and logical expressions. For example, we can write

$$\left\{ (y,z) \;\middle|\; \begin{array}{c} y \leftarrow B \\ z \leftarrow C(y) \end{array} \right\}$$

to denote the random variable $x := (y,z)$ that is obtained by the algorithm A that samples y according to B, samples z according to C on input y, and outputs the pair (y,z). We can also write

$$\Pr\left[y = z \;\middle|\; \begin{array}{c} y \leftarrow B \\ z \leftarrow C(y) \end{array} \right]$$

to denote the probability that the random variable $x = (y,z)$ sampled as above is such that $f(x) = 1$ where f is the boolean function that given a pair $x = (y,z)$ outputs 1 if and only if $y = z$.

2.2 Information theory

We recall the definition of the entropy function with respect to a set.

Definition 2.2. *For $q \in \mathbb{N}$, the q-***entropy** *function is the function $H_q \colon [0,1] \to \mathbb{R}$ defined as*

$$H_q(x) = x \cdot \log_q(q-1) - x \log_q(x) - (1-x) \cdot \log_q(1-x) \,.$$

*We overload notation, and, for a set S, let $H_S(x) := H_{|S|}(x)$ be the S-***entropy** *function.*

The (fractional) Hamming distance between two vectors is the fraction of points in which they differ. We define this notion formally as well as the restricted Hamming distance, which represents the Hamming distance over a subset of vector entries.

Definition 2.3. *Let $f, g \colon S \to \Sigma$ and let $T \subseteq S$. The* **restricted (fractional) Hamming distance** *between f and g over T is*

$$\Delta_T(f,g) = \Pr_{i \leftarrow T}[f(i) \neq g(i)] \,.$$

The **(fractional) Hamming distance** *between f and g is $\Delta(f,g) = \Delta_S(f,g)$. For a set $\mathcal{C}$, we write $\Delta_T(f,\mathcal{C}) = \min_{g \in \mathcal{C}} \Delta_T(f,g)$ and $\Delta(f,\mathcal{C}) = \min_{g \in \mathcal{C}} \Delta(f,g)$.*

The volume of the Hamming ball of distance δ around a point in space is the number of points in the space whose Hamming distance to the point is at most δ:

Definition 2.4. *For any $q, n \in \mathbb{N}$ and $\delta \in (0,1)$, we denote the* **volume of the Hamming ball** *of radius δ over space q^n is*

$$\mathsf{Vol}_q(\delta, n) = \sum_{i=0}^{\lfloor \delta \cdot n \rfloor} \binom{n}{i} \cdot (q-1)^i \,.$$

2.3 Error correcting codes

Basic definitions. We recall the definition of error correcting codes.

Definition 2.5. *An* **error correcting code** *$\mathcal{C}$ over an alphabet Σ with block length n is a subset $\mathcal{C} \subseteq \Sigma^n$. The* **minimum distance of** *$\mathcal{C}$ is $\delta_{\mathsf{min}}(\mathcal{C}) \coloneqq \min_{f,g \in \mathcal{C}} \Delta(f,g)$, and the* **rate** *of $\mathcal{C}$ is $\rho(\mathcal{C}) \coloneqq \frac{\log_{|\Sigma|} |\mathcal{C}|}{n}$.*

We interchangeably consider a code $\mathcal{C}$ as a subset $\mathcal{C} \subseteq \Sigma^n$ and as the injective map $\mathcal{C} \colon \Sigma^k \to \Sigma^n$. We also write $\mathsf{enc}_{\mathcal{C}}$ for the encoding time of $\mathcal{C}$.

A crude estimate of the tradeoff between the rate of a code $\rho(\mathcal{C})$ and its minimum distance $\delta_{\mathsf{min}}(\mathcal{C})$ is given by the Singleton bound.

Lemma 2.6. *Let $\mathcal{C} \subseteq \Sigma^n$ be an error correcting code. Then,*

$$|\mathcal{C}| \leq |\Sigma|^{n \cdot (1 - \delta_{\mathsf{min}}(\mathcal{C}) + 1/n)} \,.$$

This implies that $\rho(\mathcal{C}) \leq 1 - \delta_{\mathsf{min}}(\mathcal{C}) + 1/n$. We say that a code $\mathcal{C}$ is **maximum distance separable (MDS)** *if $\rho(\mathcal{C}) = 1 - \delta_{\mathsf{min}}(\mathcal{C}) + 1/n$.*

Most codes that we consider have additional *linear* structure.

Definition 2.7. *Let $\mathbb{F}$ be a field. An error correcting code $\mathcal{C} \subseteq \Sigma^n$ is* **$\mathbb{F}$-additive** *if $\Sigma = \mathbb{F}^s$ where $s \in \mathbb{N}$ and $\mathcal{C}$ is a $\mathbb{F}$-linear subspace of Σ^n. If $s = 1$ the code $\mathcal{C}$ is* **$\mathbb{F}$-linear***.*

Unique and list decoding. By definition, for a code $\mathcal{C} \subseteq \Sigma^n$, for every word $f \in \Sigma^n$ there is at most one codeword $u \in \mathcal{C}$ with $\Delta(f, u) < \delta_{\mathsf{min}}(\mathcal{C})/2$. For this reason, we call $\delta_{\mathsf{min}}(\mathcal{C})/2$ the ***unique-decoding*** radius. This definition can be naturally extended to consider the number of codewords within arbitrary radius δ:

Definition 2.8. *Let $\mathcal{C}$ be an error correcting code over alphabet Σ with block length n. For any $f \colon [n] \to \Sigma$ and $\delta \in (0,1)$, we define the* **list around f at distance δ** *as*

$$\Lambda(\mathcal{C}, \delta, f) \coloneqq \{ g \in \mathcal{C} \mid \Delta(f,g) \leq \delta \} \,.$$

Further, we write $|\Lambda(\mathcal{C}, \delta)| \coloneqq \max_f |\Lambda(\mathcal{C}, \delta, f)|$.

Observe that for every $\delta_1 < \delta_2$ it holds that $0 \leq |\Lambda(\mathcal{C}, \delta_1)| \leq |\Lambda(\mathcal{C}, \delta_2)| \leq |\mathcal{C}|$.

Interleaved codes. An interleaved code is constructed by bundling together multiple codewords of another code:

Definition 2.9. *Let $\mathcal{C}$ be an error correcting code over alphabet Σ with block length n, and let $m \in \mathbb{N}$. The m-**interleaved code** $\mathcal{C}^{\equiv m}$ is the code over alphabet Σ^m with block length n defined as*

$$\mathcal{C}^{\equiv m} := \{(u_1, \ldots, u_m) \mid \forall i \in [m] : u_i \in \mathcal{C}\} \subseteq (\Sigma^m)^n .$$

That is, a codeword $u \in \mathcal{C}^{\equiv m}$ consists of $m \cdot n$ symbols of Σ, arranged as an $m \times n$ size matrix where each row corresponds to a codeword of $\mathcal{C}$ (and interpreted as a vector of length n over the alphabet Σ^m).

Clearly, for any $m \in \mathbb{N}$ and $\delta \in (0, 1)$ it holds that $|\Lambda(\mathcal{C}, \delta)| \leq |\Lambda(\mathcal{C}^{\equiv m}, \delta)| \leq |\Lambda(\mathcal{C}, \delta)|^m$. In fact the list size of an interleaved code at some distance is larger than the list size of the base code by a factor *independent of the interleaving factor* (but which depends on the distance δ considered).

Lemma 2.10 ([GGR11])**.** *Let $\mathcal{C}$ be a code and consider the following: $\delta \in [0, \delta_{\mathsf{min}}(\mathcal{C}))$, $\eta := \delta_{\mathsf{min}}(\mathcal{C}) - \delta$, $b := \left\lceil \frac{\delta}{\eta} \right\rceil$, $r := \left\lceil \log \frac{\delta_{\mathsf{min}}(\mathcal{C})}{\eta} \right\rceil$, and $m \geq 1$. Then $|\Lambda(\mathcal{C}^{\equiv m}, \delta)| \leq \binom{b+r}{r} \cdot |\Lambda(\mathcal{C}, \delta)|^r$.*

2.4 Reed–Solomon codes and their variants

The main body of this paper will center around a specific error-correcting code known as the Reed–Solomon code.

Definition 2.11. *Let $\mathbb{F}$ be a finite field, $\mathcal{L} \subseteq \mathbb{F}$ and $k \in \mathbb{N}$. The* **Reed–Solomon code** *over alphabet $\mathbb{F}$ with message length k and evaluation domain $\mathcal{L}$ is defined as*

$$\mathsf{RS}[\mathbb{F}, \mathcal{L}, k] := \left\{ f \colon \mathcal{L} \to \mathbb{F} \;\middle|\; \exists \hat{f} \in \mathbb{F}^{<k}[X], \forall x \in \mathcal{L} : \; \hat{f}(x) = f(x) \right\} .$$

We let $n := |\mathcal{L}|$ denote the block length of the code and $\rho := \frac{k}{n}$ denote the rate.

Note that we interpret codewords in $\mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ as functions $f \colon \mathcal{L} \to \mathbb{F}$ by making use of the bijection from $[n]$ to $\mathcal{L}$.

The Reed–Solomon code has distance $\delta_{\mathsf{min}}(\mathcal{C}) = \frac{n-k+1}{n} = 1 - \rho + \frac{1}{n}$, achieving the Singleton bound (Lemma 2.6), and thus it is a MDS code.

The most common scenario to use Reed–Solomon codes in applications is over a domain $\mathcal{L}$ which is highly structured, called a “smooth” domain. This is in large part because, in this setting, Reed–Solomon codes have a quasi-linear time encoding via Fast-Fourier-Transforms (FFTs).

Definition 2.12. *We say that a set $\mathcal{L} \subseteq \mathbb{F}$ is* **smooth** *if it is a multiplicative coset of a subgroup $H \subseteq \mathbb{F}^*$ whose order is a power of two.*

Interleaved Reed–Solomon codes. In practice, it is quite common to use the *interleaved* Reed–Solomon code, rather than the standard one. Among other benefits, taking the s-interleaving of a Reed–Solomon code allows encoding to be done via s FFTs, each of size n/s. This is in contrast to the single FFT of size n needed for an equivalent Reed–Solomon code. Since FFTs are a quasi-linear time operation, this significantly improves concrete efficiency.

Definition 2.13. *Let $\mathbb{F}$ be a finite field, $\mathcal{L} \subseteq \mathbb{F}$ and $k, s \in \mathbb{N}$. The* **s-interleaved Reed–Solomon code** *over alphabet $\mathbb{F}$ with message length k and evaluation domain $\mathcal{L}$ is defined as*

$$\mathsf{IRS}[\mathbb{F}, \mathcal{L}, k, s] := (\mathsf{RS}[\mathbb{F}, \mathcal{L}, k/s])^{\equiv s} .$$

Folded Reed–Solomon codes. Another variation of Reed–Solomon codes which are notable for having very strong properties (e.g., list decoding and MCA up to capacity) are "folded" Reed–Solomon codes. These are constructed by combining every s consecutive symbol of a Reed–Solomon code into a single alphabet symbol.

First, we define admissible field elements. They ensure that in a codeword of a folded Reed–Solomon an evaluation point appears only once.

Definition 2.14. *Let $\mathcal{L} \subseteq \mathbb{F}$ and $s \in \mathbb{N}$. We say that $\omega \in \mathbb{F}$ is $(\mathcal{L}, s)$-**admissible** if for every $\alpha, \beta \in \binom{\mathcal{L}}{2}$ it holds that $\alpha \cdot \omega^i \neq \beta$ for every $0 \leq i < s$.*

Using this definition, we may define folded Reed–Solomon codes.

Definition 2.15 ([GR08])**.** *Let $\mathbb{F}$ be a finite field, $k, s \in \mathbb{N}$, $\mathcal{L} \subseteq \mathbb{F}$, and $\omega \in \mathbb{F}$ be $(\mathcal{L}, s)$-admissible. We define the* **folded Reed–Solomon code***:*

$$\mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega] := \left\{ f : \mathcal{L} \to \mathbb{F}^s \;\middle|\; \exists \hat{f} \in \mathbb{F}^{<k}[X],\ \forall x \in \mathcal{L} \colon f(x) = \begin{bmatrix} \hat{f}(x) \\ \hat{f}(x \cdot \omega) \\ \vdots \\ \hat{f}(x \cdot \omega^{s-1}) \end{bmatrix} \right\} .$$

From now on, we forgo specifying that ω is $(\mathcal{L}, s)$-admissible when clear from context. Note that $\mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, 1, \omega] = \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$. Further, the encoding time of the folded Reed–Solomon code is the same as that of a Reed–Solomon code of block length $n \cdot s$.

2.5 Subspace-design codes

Recent developments [CZ25; GG25] show that a large family of codes achieve desirable properties such as list decodability and mutual correlated agreement *up to capacity*. Most of these codes possess this property by virtue of satisfying an algebraic property, known as the "subspace-design" property, which we recall next.

Definition 2.16 ([GX13])**.** *A $\mathbb{F}$-additive code $\mathcal{C} \colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ is a τ-**subspace-design code** if for every $r \in \mathbb{N}$ and $\mathbb{F}$-linear subspace $\mathcal{A}$ of $\mathcal{C}$ with $\dim \mathcal{A} \leq r$ it holds that*

$$\frac{\sum_{i \in [n]} \dim \mathcal{A}_i}{n} \leq \dim \mathcal{A} \cdot \tau(r) ,$$

where $\mathcal{A}_i := \{a \in \mathcal{A} \;:\; a_i = 0^s\}$.

For any subspace-design code, the function τ is lower-bounded by $\rho - 1/n$.

Lemma 2.17 ([GG25])**.** *If $\mathcal{C} \colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ is a τ-subspace-design code of rate ρ then $\min_{r \in \mathbb{N}} \tau(r) \geq \rho - 1/n$.*

Two important examples of codes which exhibit a subspace-design structure are folded Reed–Solomon codes (see Definition 2.15) and univariate multiplicity codes. In this paper we consider two codes which are subspace-design: folded Reed–Solomon codes (see Definition 2.15) and univariate multiplicity codes (see Section A.2).

Theorem 2.18 ([GK16])**.** *Let $\mathcal{C} \colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a code with rate ρ. If $\mathcal{C}$ is:*

- *A folded Reed–Solomon code* $\mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega]$*, or*

- *A univariate multiplicity code* $\mathsf{UM}[\mathbb{F}, \mathcal{L}, k, s]$ *with* $|\mathbb{F}| > n$ *and* $\mathsf{char}(\mathbb{F}) > \rho s n > s$*.*

Then $\mathcal{C}$ *is* τ*-subspace-design for* $\tau(r) := \begin{cases} \frac{s \cdot \rho}{s-r+1} & r \in [s] \\ 1 & o.w. \end{cases}$*.*

2.6 Extension fields

We review definitions about extension fields and codes defined over extension fields.

Definition 2.19. *A tuple* $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ *is an* **extension field presentation** *if:*
- $\mathbb{B}, \mathbb{F}$ *are fields;*
- $\psi \colon \mathbb{B} \hookrightarrow \mathbb{F}$ *is an injective field homomorphism;*
- $e \in \mathbb{N}$ *is the dimension of* $\mathbb{F}$ *as a* $\mathbb{B}$*-vector space;*
- $\varphi \colon \mathbb{F} \to \mathbb{B}^e$ *is a* $\mathbb{B}$*-linear isomorphism.*

For each $i \in [e]$*,* $\varphi_i \colon \mathbb{F} \to \mathbb{B}$ *denotes the* i*-th component of* φ*. An extension field presentation* $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ *is* **systematic** *if, for every* $x \in \mathbb{B}$*,* $\varphi(\psi(x)) = (x, 0, \ldots, 0)$*.*

We also apply ψ to vectors of base field elements and $\varphi, \varphi_1, \ldots, \varphi_e$ to vectors of extension field elements by applying each of these maps componentwise.

An extension code defines a code over a field extension for a linear code over the base field.

Definition 2.20. *Let* $\mathcal{C}_{\mathbb{B}} \colon \mathbb{B}^k \to \mathbb{B}^n$ *be a linear code. The* **extension code of** $\mathcal{C}_{\mathbb{B}}$ **over** $\mathbb{F}$ *(with respect to the extension field presentation* $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$*) is the linear code* $\mathcal{C}_{\mathbb{F}} \colon \mathbb{F}^k \to \mathbb{F}^n$ *defined as*

$$\mathcal{C}_{\mathbb{F}}(\boldsymbol{v}) := \varphi^{-1}\Big(\mathcal{C}_{\mathbb{B}}(\varphi_1(\boldsymbol{v})), \ldots, \mathcal{C}_{\mathbb{B}}(\varphi_e(\boldsymbol{v}))\Big) .$$

If the extension field presentation is systematic, then $\mathcal{C}_{\mathbb{F}}(\psi(\boldsymbol{v})) = \psi(\mathcal{C}_{\mathbb{B}}(\boldsymbol{v}))$ for any $\boldsymbol{v} \in \mathbb{B}^k$.

Many properties of the extension code are closely related to the base code. For example, distance of the extension code $\mathcal{C}_{\mathbb{F}}$ equals the distance of the given code $\mathcal{C}_{\mathbb{B}}$, i.e., $\delta_{\mathsf{min}}(\mathcal{C}_{\mathbb{B}}) = \delta_{\mathsf{min}}(\mathcal{C}_{\mathbb{F}})$ (see e.g. [DP25, Thm 3.2]).

Further, the list size of the extension code is the list size of the e-interleaved base code (which in turn is related to the list size of the base code by Lemma 2.10).

Lemma 2.21. *[BCFW25, Lemma D.3] Let* $\mathcal{C}_{\mathbb{B}} \colon \mathbb{B}^k \to \mathbb{B}^n$ *be a linear code and* $(\mathbb{B}, \mathbb{F}, e, \psi, \varphi)$ *be an extension field presentation. For every* $\delta \in (0, 1)$*,* $|\Lambda(\mathcal{C}_{\mathbb{F}}, \delta)| = |\Lambda(\mathcal{C}_{\mathbb{B}}^e, \delta)|$*.*

3 List decoding

List decoding [Eli57; Woz58] is a fundamental question in the theory of error-correcting codes. The *combinatorial* list decoding question is to pin down the list size $|\Lambda(\mathcal{C}, \delta)|$ for different codes and, specifically, to understand in which regimes the list size is small. Another important avenue of research is the *algorithmic* list decoding question: given a word f and a proximity bound δ such that $|\Lambda(\mathcal{C}, \delta, f)| = \mathrm{poly}(n)$, efficiently compute the list $\Lambda(\mathcal{C}, \delta, f)$. In this paper, we focus on the combinatorial question, as most modern SNARK constructions do not require efficient list decoding.

In this section we describe relevant known results about list decoding. We note that this summary is not meant to be an exhaustive list of all known results, but rather ones that are the most relevant to the development of SNARKs. For further reading, see the recent survey of Kumar and Ron-Zewi [KR26] on list decoding of polynomial codes.

3.1 Positive results

Johnson bound. A fundamental result in the list decoding of error-correcting codes is the Johnson bound, which gives an upper bound for $|\Lambda(\mathcal{C}, \delta)|$ for any error-correcting code up to a certain distance.

Definition 3.1. *For $q, \ell \in \mathbb{N}$, we define the following three functions $J_{q,\ell}, J_q, J \colon (0,1) \to \mathbb{R}$ which are successively rougher bounds, all colloquially referred to as the* **Johnson bound***:*

- $J_{q,\ell}(\delta) = \left(1 - \frac{1}{q}\right) \cdot \left(1 - \sqrt{1 - \frac{q}{q-1} \cdot \frac{\ell}{\ell-1} \cdot \delta}\right)$.
- $J_q(\delta) = \lim_{\ell\to\infty} J_{q,\ell}(\delta) = \left(1 - \frac{1}{q}\right) \cdot \left(1 - \sqrt{1 - \frac{q}{q-1} \cdot \delta}\right)$.
- $J(\delta) = \lim_{q\to\infty} J_q(\delta) = 1 - \sqrt{1-\delta}$.

Theorem 3.2 ([Joh62]). *For any $\mathcal{C} \subseteq \Sigma^n$ with $|\Sigma| = q$ it holds that $|\Lambda(\mathcal{C}, J_{q,\ell}(\delta_{\mathsf{min}}(\mathcal{C})))| \leq \ell$.*

Recalling that MDS codes (which include the important class of interleaved Reed–Solomon codes) have minimum relative distance bounded by $1 - \rho$, we can derive the following useful coarse list decoding bound:

Corollary 3.3. *For every MDS code $\mathcal{C}$ with rate ρ and every $\eta > 0$: $\left|\Lambda\left(\mathcal{C}, 1 - \sqrt{\rho} - \eta\right)\right| \leq \frac{1}{2\cdot\eta\cdot\rho}$.*

While the Johnson bound is a combinatorial statement showing that the list size can be polynomial for a large parameter regime, it is not generally computationally feasible to find this list. However, in the case of (generalized) Reed–Solomon codes, the celebrated Guruswami–Sudan list decoding algorithm [GS99] finds, for every word f, the list $\Lambda(\mathcal{C}, 1 - \sqrt{\rho} - \eta, f)$ in polynomial time provided we are in the range where $|\Lambda(\mathcal{C}, 1 - \sqrt{\rho} - \eta)|$ is polynomial (recently, it was shown that this can even be done deterministically [CHK25]). Thus, for Reed–Solomon codes up to the Johnson bound, the combinatorial and algorithmic list decoding bounds match.

Beyond the Johnson bound. For general codes (or even linear codes) it is known that the Johnson bound is *tight* (e.g., [Gur02; GS03]). However, there are specific classes of codes for which non-trivial list decoding bounds are known beyond the Johnson bound and even up to capacity.

An important example for this paper are subspace-design codes:

Theorem 3.4 ([CZ25], Theorem B.5)**.** *Let $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a τ-subspace-design code. For every $\eta > 0$ it holds that*

$$|\Lambda(\mathcal{C}, 1-\tau(1/\eta)-\eta)| \le \frac{1-\tau(1/\eta)}{\eta}\,.$$

Since folded Reed–Solomon codes are subspace-design codes (see Theorem 2.18) the above theorem implies that they are list decodable up to capacity (a similar corollary can be derived for univariate multiplicity codes):

Corollary 3.5 ([CZ25], Corollary 2.21)**.** *Let $\mathcal{C} = \mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega]$ be a folded Reed–Solomon code of rate ρ. Then, for any $\eta > 0$ such that $1/\eta < s$ it holds that*

$$\left|\Lambda\left(\mathcal{C}, 1 - \frac{\rho \cdot s}{s - 1/\eta + 1} - \eta\right)\right| \le \frac{s\cdot(1-\rho) + 1 - \frac{1}{\eta}}{\eta\left(s + 1 - \frac{1}{\eta}\right)}\,.$$

Observe that if $\eta \ge \sqrt{3/s}$ then $|\Lambda(\mathcal{C}, 1-\rho-\eta)| \le \frac{1}{\eta}$.

In recent years, an exciting line of work progressed toward showing combinatorial list decoding of Reed–Solomon codes close to capacity, with the caveat that these codes are obtained by *randomly* choosing the evaluation domain.

Over an exponentially sized alphabet [BGM23] showed that with high probability these codes are list decodable up to a distance $1-\rho-\eta$ with list size $\frac{1-\rho-\eta}{\eta}$, resolving the conjecture posed in [ST20]. The result was later improved by [GZ23] to polynomial sized alphabet, and further strengthened in [AGL24] which shows the following result (see also [AGGLZ25] which combines the two prior works).

Theorem 3.6 ([AGL24], Theorem 1.1)**.** *Let $\ell \ge 2$, $\eta \in (0,1)$, $k \in \mathbb{N}$, $n \in \mathbb{N}$ and let $\mathbb{F}$ be a finite field with $|\mathbb{F}| \ge n + k \cdot 2^{10\ell/\eta}$. Then,*

$$\Pr\left[\left|\Lambda\left(\mathcal{C}, \frac{\ell}{\ell+1}\cdot(1-\rho-\eta)\right)\right| \le \ell \;\middle|\; \begin{matrix} \mathcal{L} \leftarrow \binom{\mathbb{F}}{n} \\ \mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, \mathcal{L}, k], \rho \coloneqq k/n \end{matrix}\right] \ge 1 - 2^{-\ell n}\,.$$

As a result, by choosing $\ell = \frac{2\cdot(1-\rho-\eta)}{\eta}$ and $\mathbb{F}$ with $|\mathbb{F}| \ge n + k \cdot 2^{20(1-\rho-\eta)/\eta^2}$, we have that with probability $1 - 2^{-2n(1-\rho-\eta)/\eta}$, the code $\mathcal{C}$ has

$$|\Lambda(\mathcal{C}, 1-\rho-\eta)| \le \frac{2\cdot(1-\rho-\eta)}{\eta}\,.$$

A similar statement, also due to [AGL24], shows that random linear codes are list decodable up to capacity with high probability.

3.2 Limitations

In this section we describe known lower bounds on the list size for error correcting codes, starting with results on general (linear) codes, and then focusing on the special case of Reed–Solomon codes.

General linear codes. The earliest lower bound to list decoding goes back to the introduction of list decoding by [Eli57], which showed that for any code the list size at a distance δ is lower-bounded by the total volume of the Hamming balls of radius δ around codewords, divided by the total number of words.

Lemma 3.7 ([Eli57])**.** *Let $\mathcal{C}\colon \Sigma^k \to \Sigma^n$ be a code with $q := |\Sigma|$. For any $\delta \in (0,1)$ it holds that*

$$|\Lambda\left(\mathcal{C},\delta\right)| \geq \frac{\mathsf{Vol}_q(\delta,n)}{q^{n-k}}\,.$$

Proof. We begin by observing that there must exist g with $|\Lambda(\mathcal{C},\delta,g)| \geq E_{f\leftarrow\Sigma^n}\left[|\Lambda(\mathcal{C},\delta,f)|\right]$, and so $|\Lambda\left(\mathcal{C},\delta\right)| \geq E_{f\leftarrow\Sigma^n}\left[|\Lambda(\mathcal{C},\delta,f)|\right]$. Finally,

$$\mathbb{E}_{f\leftarrow\Sigma^n}\left[|\Lambda(\mathcal{C},\delta,f)|\right] = \sum_{u\in\mathcal{C}} \Pr_{f\leftarrow\Sigma^n}[\Delta(f,u)\leq\delta] = \sum_{u\in\mathcal{C}} \frac{\mathsf{Vol}_q(\delta,n)}{q^n} = \frac{\mathsf{Vol}_q(\delta,n)}{q^{n-k}}\,.$$

□

The volume of a Hamming ball can be roughly estimated by $\mathsf{Vol}_q(\delta,n) \approx q^{n\cdot(\rho-1+H_q(\delta))}$. Using a more precise lower-bound on $\mathsf{Vol}_q(\delta,n)$ from [MS77] we derive the following:[4]

Corollary 3.8. *Let $\mathcal{C}\colon \Sigma^k \to \Sigma^n$ be a code of rate ρ and let $q := |\Sigma|$. For any $\delta \in (0,1)$,*

$$|\Lambda\left(\mathcal{C},\delta\right)| \geq \frac{q^{n\cdot(\rho-1+H_q(\delta))}}{\sqrt{8\cdot n\cdot\delta\cdot(1-\delta)}}\,.$$

The generalized Singleton bound generalizes Singleton's classical bound [Sin64] to list decoding (here we focus on the bound for linear codes):

Theorem 3.9 ([ST20], Theorem 1.2)**.** *Let $\mathbb{F}$ be finite field, $0 < \ell < |\mathbb{F}|$, $\delta \in (0,1)$ and let $\mathcal{C}\colon \mathbb{F}^k \to \mathbb{F}^n$ be a linear error correcting code of rate ρ. If $|\Lambda\left(\mathcal{C},\delta\right)| \leq \ell$, then*

$$|\mathcal{C}| \leq |\mathbb{F}|^{n-\lfloor\frac{\ell+1}{\ell}\cdot\delta\cdot n\rfloor}\,.$$

Consequently, $\delta \leq \frac{\ell}{\ell+1}\cdot(1-\rho)$.

We recall a limitation on achieving the generalized singleton bound, first shown in [BDG24, Corollary 1.7, Thm 1.8] for the case when $\ell = 2$ and then generalized in [AGL23] to the following result.

Theorem 3.10 ([BDG24; AGL23])**.** *Let $\ell \geq 2$ and $\rho \in (0,1)$ be constants. There exists a constant $\alpha_{\ell,\rho}$ such that for any $\eta > 0$ and any sufficiently large $n \geq \Omega_{\ell,\rho}(1/\eta)$ any error correcting code $\mathcal{C}\colon \Sigma^k \to \Sigma^n$ of rate ρ with $\left|\Lambda\left(\mathcal{C},\frac{\ell}{\ell+1}\cdot(1-\rho-\eta)\right)\right| \leq \ell$ has $|\Sigma| \geq 2^{\alpha_{\ell,\rho}/\eta}$.*

In particular, this shows that achieving *exactly* the generalized singleton bound (which implies the case when $\eta = \Theta(1/n)$) requires an alphabet of exponential size, which is undesirable.

While random linear codes have many desirable properties, limitations to their list size have also been shown:

[4] See [DG25] for further analysis bounding this value.

Theorem 3.11 ([GLMRSW22], Theorem 4.1)**.** *Fix a prime power q, fix $\delta \in \left(0, 1 - \frac{1}{q}\right)$, and fix $\varepsilon \in (0,1)$. There exists $\gamma > 0$ such that for all $1 - h_q(\delta) - \gamma < \rho < 1 - h_q(\delta)$ and all sufficiently large n, a uniformly random linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$ of rate ρ satisfies*

$$|\Lambda(\mathcal{C}, \delta)| \; > \; \left\lfloor \frac{h_q(\delta)}{1 - h_q(\delta) - \rho} - \varepsilon \right\rfloor$$

with probability $1 - q^{-\Omega(n)}$.

Reed–Solomon codes. As Reed–Solomon codes and their variants are of special interest to this paper, we give further limitations specific to this class:

Theorem 3.12 ([BKR06], Corollary 2.2)**.** *Fix $0 < \alpha < \beta < 1$. For infinitely many (prime powers) q, there exists $w : \mathbb{F}_q \to \mathbb{F}_q$ such that for $\mathcal{C} := \mathsf{RS}[\mathbb{F}_q, \mathbb{F}_q, \lfloor q^\alpha \rfloor]$:*

$$|\Lambda(\mathcal{C}, 1 - q^{\beta - 1}, w)| \geq q^{(\alpha - \beta^2)\log q} \; = \; 2^{(\alpha - \beta^2)(\log q)^2} \, .$$

In other words, there are codes where the list size is superpolynomial in the block-length. Subsequently, there can be no efficient list decoder for such Reed–Solomon codes.

Theorem 3.12 relies heavily on extension-field structure, and therefore does not directly apply to prime fields. For prime fields, the best-known bounds are split into two regimes. The following theorem deals with the same low-rate regime as Theorem 3.12, but applies to prime fields, achieving slightly worse distances but significantly larger list-sizes:

Theorem 3.13 ([GHSZ02], Corollary 20)**.** *Fix $0 < \alpha, \beta < 1$. For all sufficiently large primes p, there exist $\mathcal{C} := \mathsf{RS}[\mathbb{F}_p, \mathbb{F}_p, \lfloor p^\alpha \rfloor]$ and a word $w \colon \mathbb{F}_p \to \mathbb{F}_p$ such that*

$$\left| \Lambda\left(\mathcal{C}, 1 - \frac{1 - \beta}{\alpha} \cdot p^{\alpha - 1}, w\right) \right| > \Omega\left(p^{p^{\alpha} \cdot \beta / 2}\right) \, .$$

On the other end of the spectrum, in the large-rate regime, further bounds are known for prime fields (the statement holds for prime powers, but applies to primes as well):

Theorem 3.14 ([JH01], Theorem 2)**.** *Fix an integer $j \geq 2$. For infinitely many prime powers q (satisfying $q = 1 \pmod{j+1}$), there exists $\mathcal{C} := \mathsf{RS}[\mathbb{F}_q, \mathcal{L}, k]$ with $|\mathcal{L}| = j + 1$ and rate $\rho \approx \frac{j-1}{j+1}$ and a word $w : \mathcal{L} \to \mathbb{F}_q$ such that:*

$$\left| \Lambda\left(\mathcal{C}, \frac{1}{j+1}, w\right) \right| > j \, .$$

Finally, we reconsider the computational list-decoding problem. While Theorem 3.6 posits that most Reed–Solomon codes have list sizes beyond the Johnson bound, efficient decoding beyond this bound is unknown. The following result shows limitations to algorithmic list decoding beyond the Johnson bound based on the cryptographic hardness of the discrete logarithm problem. Roughly, the discrete logarithm problem[5] in a field $\mathbb{F}$ is as follows: given two elements $g, h \in \mathbb{F}^*$ as input, where h is an element in the subgroup of $\mathbb{F}^*$ generated by g, find x such that $h = g^x$.

[5] See [CW07] for a more formal definition of the discrete logarithm problem in the context of Reed–Solomon codes.

Theorem 3.15 ([CW07]). *Fix a prime power q and consider a Reed–Solomon code $\mathcal{C} := \mathsf{RS}[\mathbb{F}_q, \mathcal{L}, k]$ where $|\mathcal{L}| = n$. Let t be the smallest integer such that $\binom{n}{t} / q^{t-k} \leq 1$ and let $\delta := 1 - t/n$. If there is a polynomial-time algorithm that, given any received word $f \in \mathbb{F}^n$, outputs all codewords of the list $\Lambda(\mathcal{C}, \delta, f)$, then one obtains a (randomized) polynomial-time algorithm for the discrete logarithm problem in $\mathbb{F}_{q^{t-k}}$.*

We stress that, generally, modern SNARK constructions do not require efficient list decoding, and so the above result is not a major obstruction in the context of this paper.

4 Correlated agreement conjectures

4.1 Definitions

Before defining correlated agreement, let us first start with a weaker notion called a *proximity gap*. Let $f_1, f_2 \in \mathbb{F}^n$ be two words, and let $L := \{f_1 + \gamma \cdot f_2\}_{\gamma \in \mathbb{F}}$ be the corresponding line in $\mathbb{F}^n$. For $\delta \in (0,1)$, let $p(L) := \Pr_{g \leftarrow L}\big[\Delta(g, \mathcal{C}) \leq \delta\big]$ be the fraction of points on L that are δ-close to the code $\mathcal{C}$. We say that a linear code $\mathcal{C}$ has a proximity gap $\varepsilon_{\mathsf{pg}}(\mathcal{C}, \delta)$ if for all $f_1, f_2 \in \mathbb{F}^n$, either $p(L) \leq \varepsilon_{\mathsf{pg}}(\mathcal{C}, \delta)$ or $p(L) = 1$. In other words, whenever $p(L) > \varepsilon_{\mathsf{pg}}(\mathcal{C}, \delta)$ then $p(L) = 1$. One can generalize beyond lines and consider proximity gaps with respect to affine spaces of bounded dimension, algebraic curves in $\mathbb{F}^n$, and more. Here, for simplicity, we will only consider proximity gaps with respect to lines.

While they are an important notion, proximity gaps do not seem very useful when constructing SNARKs, and will not be focused upon in this section.

Correlated agreement. Correlated agreement (CA) refines the notion of a proximity gap. We say that a linear code $\mathcal{C}$ has CA error $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$ if for all $f_1, f_2 \in \mathbb{F}^n$, whenever $p(L) > \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$ we have that $\Delta\big((f_1, f_2),\ \mathcal{C}^{\equiv 2}\big) \leq \delta$. In other words, whenever $p(L) > \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$, not only are f_1, f_2 close to $\mathcal{C}$, but there is a set $S \subseteq [n]$ where $|S| \geq (1-\delta)n$, such that f_1 and f_2 match two codewords in $\mathcal{C}$ at all the positions in this set S. It is not difficult to see that this implies that $p(L) = 1$. Because correlated agreement implies a proximity gap, it follows that $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta) \geq \varepsilon_{\mathsf{pg}}(\mathcal{C}, \delta)$ for all codes $\mathcal{C}$ and all $\delta \in (0,1)$. The following definition captures $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$ more precisely, generalizes to $\mathbb{F}$-additive codes and allows for loss in proximity.

Definition 4.1. *The* **correlated agreement (CA) error** *of an $\mathbb{F}$-additive code $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ with respect to distances $\delta_{\mathsf{fld}}, \delta_{\mathsf{int}} \in (0,1)$ (with $\delta_{\mathsf{fld}} \leq \delta_{\mathsf{int}}$) is*

$$\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}, \delta_{\mathsf{int}}) := \max_{f_1, f_2 \in (\mathbb{F}^s)^n} \left\{ \Pr_{\gamma \leftarrow \mathbb{F}} \begin{bmatrix} \Delta(f_1 + \gamma \cdot f_2,\ \mathcal{C}) \leq \delta_{\mathsf{fld}} & \wedge \\ \Delta\big((f_1, f_2),\ \mathcal{C}^{\equiv 2}\big) > \delta_{\mathsf{int}} & \end{bmatrix} \right\}.$$

The quantity $\epsilon = \delta_{\mathsf{int}} - \delta_{\mathsf{fld}}$ is called the **proximity loss***. When $\epsilon = 0$ we write $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$ (where $\delta \in (0,1)$).*

Remark 4.2. For every $f, g \in (\mathbb{F}^s)^n$ it holds that $\Delta(f, g) \in \left\{0, \frac{1}{n}, \ldots, \frac{n-1}{n}, 1\right\}$. As a result, for any $\beta, \beta' \in [0, 1/n)$ we have $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta, \delta + \beta) = \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta, \delta + \beta') = \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta)$.

Mutual correlated agreement. Mutual correlated agreement (MCA) further refines the notion of correlated agreement. Let $f_1, f_2 \in \mathbb{F}^n$ and consider again the line $L := \{f_1 + \gamma \cdot f_2\}_{\gamma \in \mathbb{F}}$. We say that the linear code $\mathcal{C}$ has MCA if for all $f_1, f_2 \in \mathbb{F}^n$, almost all points on L satisfy the following condition: if $g = f_1 + \gamma \cdot f_2$ on L satisfies $\Delta_{S_\gamma}(g, \mathcal{C}) = 0$ for some $S_\gamma \subseteq [n]$ of size at least $(1-\delta)n$ (meaning that g is δ-close to $\mathcal{C}$) then $\Delta_{S_\gamma}\big((f_1, f_2),\ \mathcal{C}^{\equiv 2}\big) = 0$. In other words, not only are f_1, f_2 mutually correlated with $\mathcal{C}$ as in CA, they must be mutually correlated with respect to the same set S_γ as the point g. This is formalized in the following definition.

Definition 4.3. *The* **mutual correlated agreement (MCA) error** *of an $\mathbb{F}$-additive code $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ with respect to distance $\delta \in (0,1)$ is*

$$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) := \max_{f_1, f_2 \in (\mathbb{F}^s)^n} \left\{ \Pr_{\gamma \leftarrow \mathbb{F}} \begin{bmatrix} \exists\, S = S_\gamma \subseteq [n] \ s.t. & : & \Delta_S(f_1 + \gamma \cdot f_2, \mathcal{C}) = 0 \ \wedge \\ |S| \geq (1-\delta) \cdot n & & \Delta_S((f_1, f_2), \mathcal{C}^{\equiv 2}) > 0 \end{bmatrix} \right\}.$$

Remark 4.4. While it is conceivable to define a version of mutual correlated agreement with proximity loss, analogous to correlated agreement, this remains to be thoroughly explored, and so we do not attempt to define such a notion in this paper.

Since MCA implies CA, which in turn implies proximity gaps, we have the following simple fact:

Fact 4.5. *For every $\mathbb{F}$-additive code $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ and distance $\delta \in (0,1)$:*

$$\varepsilon_{\mathsf{pg}}(\mathcal{C},\delta) \leq \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) \leq \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta).$$

In the other direction, it is unknown whether having a proximity gap implies CA or whether CA implies MCA. A simple result towards shows that, at least in the unique decoding regime, CA and MCA are equivalent:

Lemma 4.6 ([ACFY25], Lemma 4.10)**.** *For any $\mathbb{F}$-additive code $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ and $\delta < \delta_{\mathsf{min}}(\mathcal{C})/2$ it holds that $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta) = \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$.*[6]

The following lemma shows that interleaving a code degrades the MCA error by at most the number of interleavings. It is an open question whether this bound is tight or can be improved.

Lemma 4.7. *For any $\mathbb{F}$-additive code $\mathcal{C} \subseteq (\mathbb{F}^s)^n$, $t \in \mathbb{N}$ and $\delta \in (0,1)$: $\varepsilon_{\mathsf{mca}}(\mathcal{C}^{\equiv t},\delta) \leq t \cdot \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$.*

4.2 Positive results

We describe the positive results of correlated agreement and, when known, mutual correlated agreement. We begin by describing results in the unique decoding regime ($\delta < \delta_{\mathsf{min}}(\mathcal{C})/2$) and then move to list decoding ($\delta \geq \delta_{\mathsf{min}}(\mathcal{C})/2$).

4.2.1 Unique decoding

We describe known results on CA and MCA in the unique-decoding regime, when $\delta < \delta_{\mathsf{min}}(\mathcal{C})/2$. While all of the results described in this section were originally shown for CA, by Lemma 4.6, whenever they hold with 0 proximity loss we are able to describe them as bounds on MCA.

For general linear codes, the best error bounds within unique decoding are known up to distance $\delta := \delta_{\mathsf{min}}(\mathcal{C})/3$:

Theorem 4.8 ([AHIV17], Appendix A)**.** *For any linear error correcting code $\mathcal{C}$ and $\delta \leq \frac{\delta_{\mathsf{min}}(\mathcal{C})}{3}$,*

$$\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta) = \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) \leq \frac{n \cdot \delta + 1}{|\mathbb{F}|}\,.$$

For the specific case of Reed–Solomon codes results are known all the way up to the unique-decoding bound:

Theorem 4.9. *Let $\mathcal{C} := \mathsf{RS}[\mathbb{F},\mathcal{L},k]$ be a Reed–Solomon code with rate ρ. Then, for $\delta_{\mathsf{fld}} \leq \frac{1-\rho}{2} < \delta_{\mathsf{min}}(\mathcal{C})/2$:*

1. *[BCIKS20, Theorem 1.4]: $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta_{\mathsf{fld}}) = \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta_{\mathsf{fld}}) \leq \frac{n}{|\mathbb{F}|}$.*

[6] The proof of [ACFY25] considers linear codes, however an identical proof holds for $\mathbb{F}$-additive codes.

2. [BCHKS25, Theorem 1.3]: for $\delta_{\mathsf{min}}(\mathcal{C})/3 \leq \delta_{\mathsf{fld}} < \delta_{\mathsf{int}}$,

$$\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}, \delta_{\mathsf{int}}) \leq \max\left\{\frac{1-\rho-\delta_{\mathsf{fld}}}{\delta_{\mathsf{fld}}(1-\rho-2\delta_{\mathsf{fld}})|\mathbb{F}|}, \frac{\delta_{\mathsf{int}}}{(\delta_{\mathsf{int}}-\delta_{\mathsf{fld}})|\mathbb{F}|}\right\} .$$

Remark 4.10. Recall, by Remark 4.2, that if $\delta_{\mathsf{int}} - \delta_{\mathsf{fld}} < 1/n$, then $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}) = \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}, \delta_{\mathsf{int}})$. Thus, in this case, in Item 2, we can set $\delta_{\mathsf{int}} - \delta_{\mathsf{fld}} = \gamma/n$ for any $\gamma < 1$, deriving:

$$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta_{\mathsf{fld}}) = \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}) = \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}}, \delta_{\mathsf{fld}} + \gamma/n) \leq \max\left\{\frac{1-\rho-\delta_{\mathsf{fld}}}{\delta_{\mathsf{fld}}(1-\rho-2\delta_{\mathsf{fld}})|\mathbb{F}|}, \frac{n\cdot\delta_{\mathsf{fld}}+\gamma}{\gamma|\mathbb{F}|}\right\} .$$

By setting $\gamma > \frac{n\cdot\delta_{\mathsf{fld}}}{n-1}$ (which is possible whenever $\delta_{\mathsf{fld}} < 1 - 1/n$), we have $\frac{n\cdot\delta_{\mathsf{fld}}+\gamma}{\gamma|\mathbb{F}|} < \frac{n}{|\mathbb{F}|}$. As a result, Item 2 presents a stronger bound than Item 1 whenever

$$\frac{1-\rho-\delta_{\mathsf{fld}}}{\delta_{\mathsf{fld}}(1-\rho-2\delta_{\mathsf{fld}})|\mathbb{F}|} < \frac{n}{|\mathbb{F}|} .$$

4.2.2 List decoding

General linear codes. In the list decoding regime (shown first by [BKS18] and improved by [BGKS20; Zei24; GKL24; BCGM25]) every linear error correcting code has mutual correlated agreement up to the "1.5 Johnson bound" i.e., whenever $\delta < 1 - \sqrt[3]{1-\delta_{\mathsf{min}}(\mathcal{C})}$.

Theorem 4.11. *For any linear error correcting code $\mathcal{C} \subseteq \mathbb{F}^n$, $\eta > 0$ and $\delta \leq 1 - \sqrt[3]{1-\delta_{\mathsf{min}}(\mathcal{C})+\eta}$,*

- *[GKL24, Theorem 3]:*

$$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \leq \left(\frac{n+6}{\eta} + \frac{2}{\eta(\sqrt[3]{1-\delta_{\mathsf{min}}(\mathcal{C})+\eta} - \sqrt{1-\delta_{\mathsf{min}}(\mathcal{C})+\eta})}\right) \cdot \frac{1}{|\mathbb{F}|} = O\left(\frac{n}{\eta\cdot|\mathbb{F}|}\right) .$$

- *[BGKS20, Lemma 3.2]: $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta_{\mathsf{fld}} = \delta, \delta_{\mathsf{int}} = \delta+\eta) \leq \frac{2}{\eta^2|\mathbb{F}|}$.*

Moreover, it was shown by Bordage, Chiesa, Guan, and Manzur [BCGM25] that MCA holds for all linear codes even in a generalization the MCA experiment where rather than looking at $f_0 + \gamma \cdot f_1$, we consider $f_0 + G(\gamma) \cdot f_1$ for a large class of functions G called "polynomial generators" (as with most of the results in this paper, this can be extended to multiple functions $f_1, \ldots, f_\ell$ and $\sum_{i=1}^{\ell} G_i(\gamma) \cdot f_i$).

Reed–Solomon codes. In the specific case of Reed–Solomon codes, stronger results are known, starting from [BKS18] who showed results up to the "double-Johnson" bound and improved by [BCIKS20] to hold up to the Johnson bound $1-\sqrt{\rho}$. The error parameters of the latter result were later improved in [BCHKS25; Hab25].

Theorem 4.12 ([BCHKS25], Theorem 4.6)**.** *Let $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ be a Reed–Solomon code with rate ρ and $\eta > 0$. Then, letting $\rho_+ = \rho + 1/n$, for $\delta < 1 - \sqrt{\rho_+} - \eta$,*

$$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \leq \frac{1}{|\mathbb{F}|} \cdot \left(\frac{2(m+\frac{1}{2})^5 + 3(m+\frac{1}{2})\delta\rho_+}{3\rho_+^{3/2}} \cdot n + \frac{m+\frac{1}{2}}{\rho_+^{1/2}}\right) = O_\rho\left(\frac{n}{\eta^5|\mathbb{F}|}\right) ,$$

where $m = \max\left\{\left\lceil\frac{\sqrt{\rho_+}}{2\eta}\right\rceil, 3\right\}$.

Subspace-design codes. Goyal and Guruswami [GG25] show that a wide variety of error correcting codes exhibit MCA up to capacity.[7] Their main result shows that any code that has sufficiently good "subspace-design properties" exhibits MCA up to capacity.

Theorem 4.13 ([GG25], Corollary 4.9)**.** *Let $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a τ-subspace-design code. Then for $t \in \mathbb{N}$*

$$\varepsilon_{\mathsf{mca}}\left(\mathcal{C}, 1 - \tau(t+1) - \frac{3}{2t}\right) \leq \frac{t \cdot n + 4t^2}{|\mathbb{F}|} .$$

While many codes can be cast as subspace-design codes, we specifically highlight folded Reed–Solomon codes as we will use them later on, in Section 6. As they are subspace-design, a similar statement can be given for univariate multiplicity codes.

Theorem 4.14 ([GG25], Corollary 4.10)**.** *Let $\eta \in (0,1)$ be a parameter and let $\mathcal{C} := \mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega]$ be a folded Reed–Solomon code with $s > 16\eta^{-2}$. Then*

$$\varepsilon_{\mathsf{mca}}\left(\mathcal{C}, 1 - \rho - \eta\right) \leq \frac{2n}{\eta|\mathbb{F}|} + \frac{24}{\eta^3|\mathbb{F}|} = O\left(\frac{n}{\eta|\mathbb{F}|} + \frac{1}{\eta^3|\mathbb{F}|}\right) .$$

A useful property of subspace-design codes is that certain "local properties" of them are shared with random Reed–Solomon codes. However, interestingly, MCA does not seem to be such a local property. While, due to this fact, we cannot use these connections in a black-box manner, Guruswami and Goyal are able to derive implications for random Reed–Solomon code by considering a different decodability notion. See [GG25] for details. These results are highly interesting in theory and as evidence for Reed–Solomon codes having MCA, but do not seem to be directly useful in practice.

Theorem 4.15 ([GG25], Theorem 5.15)**.** *There exists a constant c_1 such that for every $n \in \mathbb{N}$, $\eta \in (0,1)$, and $\rho \in (0,1)$ with $\eta > c_1 \cdot n^{-1/9}$ there exists c_2 such that for every prime field $\mathbb{F} = \mathbb{F}_p$ with $p > c_2 \cdot n$ the following holds:*

$$\Pr\left[\varepsilon_{\mathsf{mca}}(\mathcal{C}, 1 - \rho - \eta) \leq \frac{2n(1-\rho) + O(1/\eta^4)}{\eta|\mathbb{F}|} \;\middle|\; \begin{array}{r} \mathcal{L} \leftarrow \binom{\mathbb{F}}{n} \\ \mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, \rho n] \end{array}\right] \geq \frac{2}{3} ,$$

where $\mathcal{L} \leftarrow \binom{\mathbb{F}}{n}$ denotes uniformly sampling a subset of size n from $\mathbb{F}$.

A similar statement with smaller MCA error term holds for general (random) linear codes.

4.3 Limitations

In this section we describe known limitations of CA and MCA. As the main focus of this paper is on MCA beyond the Johnson bound, we will primarily discuss this parameter regime. However, there is a significant body of research into understanding the exact behavior of CA and MCA within the unique-decoding radius (see e.g, [BGKS20; BCIKS20; BCHKS25]).

Error increase on approaching capacity. Our main interest is pushing CA as close as possible to the capacity regime. Roughly, the following theorem shows that one cannot hope to prove a general result for CA up to distance $1/\log n$ from capacity with error that is polynomial in n. A variant of this lower-bound was shown to hold in [BCHKS25] under a conjecture, which was later proven and improved upon by [KK25].

[7] A subset of the results of [GG25] were independently discovered by Jeronimo, Liu and Rajpal [JLR26].

Theorem 4.16 ([BCHKS25; KK25])**.** *For every $c > 0$ and rate $\rho \in (0, 1/2)$ there exists a power of two $n \in \mathbb{N}$ and Reed–Solomon code $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ of rate ρ where $|\mathbb{F}| = \mathrm{poly}(n)$ is prime and $\mathcal{L}$ is smooth with $|\mathcal{L}| = n$ such that:*

$$\varepsilon_{\mathsf{ca}}\left(\mathcal{C}, 1 - \rho - \Theta\left(\frac{1}{\log n}\right)\right) \geq \frac{n^c}{|\mathbb{F}|} .$$

Notably, the theorem above captures both prime fields and smooth domains. Thus it shows that for parameters where CA is used in practice we must have the distance from capacity be $\Omega(1/\log n)$ (at least in order to make general asymptotic statements).

The following theorem shows that at some point even this "super-polynomial" behavior of CA completely breaks down:

Theorem 4.17 ([CS25], Corollary 1)**.** *Let $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ with $q = |\mathbb{F}| \geq 10$ and rate ρ and δ be such that*

$$1 - H_q(\delta) + \frac{2}{n} + \sqrt{\frac{H_q(\delta) - \delta}{n}} \leq \rho \leq 1 - \delta - \frac{2}{n},$$

then $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta) = 1$.

To interpret the above theorem, observe that, as q grows, we have $H_q(\delta) \approx \delta - 1/\log q$. Thus, reordering the theorem in terms of δ, we have that CA error breaks down at distances $1 - \rho - \eta$ for $\eta \approx 1/\sqrt{n \log q} + 2/n - 1/\log q$.

Error jump at the Johnson bound. Theorem 4.12 shows that when approaching the Johnson bound, the error follows roughly $n/|\mathbb{F}|$. That is, at a proximity of $J(\delta_{\mathsf{min}}(\mathcal{C})) - \eta$, the MCA error is at most $O_\rho\left(\frac{n}{\eta^5 |\mathbb{F}|}\right)$. The following theorem shows that when we are exactly on the Johnson bound, the CA (and therefore MCA) error must be at least $\Omega(n^2/|\mathbb{F}|)$.

Theorem 4.18 ([BCHKS25], Corollary 1.7)**.** *Fix a constant $\epsilon > 0$ and let $\delta = 15/16$. Then for all $\mathbb{F}$ of characteristic 2 there exists a code $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ with $n = |\mathbb{F}|^{\frac{1}{2}(1+\epsilon)}$ and minimum distance $\delta_{\mathsf{min}}(\mathcal{C}) = 15/16$ such that*

$$\varepsilon_{\mathsf{ca}}\left(\mathcal{C}, \delta_{\mathsf{fld}} = J(\delta_{\mathsf{min}}(\mathcal{C})), \delta_{\mathsf{int}} = J(\delta_{\mathsf{min}}(\mathcal{C})) + \frac{1}{8} + \frac{1}{n}\right) \geq \frac{n^{2(1-\epsilon)}}{|\mathbb{F}|} .$$

Note that the above theorem holds for fields of characteristic 2. Thus it may still be the case that MCA holds with small error beyond the Johnson bound for our main point of interest, which is prime fields over smooth domains.

CA error is bounded by sampling probability. An intriguing fact, used by both [DG25] and [CS25] to achieve lower-bounds for CA error, is that the CA error of error correcting codes at any distance δ is bounded from below by (roughly) the probability that a random word is δ-close to the code. Thus, codes where the latter probability is large must have large CA error.

Lemma 4.19 ([DG25], Theorem 2.5)**.** *Let $\mathcal{C} \subseteq \mathbb{F}^n$ be a linear code and let $\delta' = \max_{u \in \mathbb{F}^n}\{\Delta(u, \mathcal{C})\}$ be the covering radius of $\mathcal{C}$. For every $\delta \in (0, \delta')$ it holds that*

$$\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta) \geq \frac{q-1}{q} \cdot \Pr_{\boldsymbol{u} \leftarrow \mathbb{F}^n}\left[\Delta(\boldsymbol{u}, \mathcal{C}) \leq \delta\right] .$$

4.4 Line-decoding

Goyal and Guruswami [GG25] prove their results on mutual correlated agreement for subspace-design codes by considering a stronger notion called line-decoding (a similar notion called "collinearity of proximates" was independently considered by Haböck [Hab24; Hab25]).

Definition 4.20 ([GG25], Definition 3.1)**.** *Let $\mathbb{F}$ be a finite field and $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ be an* $\mathbb{F}$-additive *code. The code $\mathcal{C}$ is (δ, a, b)* **line-decodable** *if for every $f_1, f_2 \in \Sigma^n$ and every function $U\colon \mathbb{F} \to \mathcal{C}$, whenever*

$$\Pr_{\gamma \leftarrow \mathbb{F}}\left[\Delta\left(f_1 + \gamma \cdot f_2,\ U(\gamma)\right) \le \delta\right] \ge \frac{a}{|\mathbb{F}|}$$

there exist $u_1, u_2 \in \mathcal{C}$ such that $\Pr_{\gamma \leftarrow \mathbb{F}}\left[U(\gamma) = u_1 + \gamma \cdot u_2\right] \ge \frac{b}{|\mathbb{F}|}$.

The above notion can be naturally generalized to general curves, in which case we call it curve-decoding. Line decodability appears implicitly in many of the proofs above as a step towards to proving CA and MCA. Indeed, MCA is directly implied by line decoding:

Theorem 4.21 ([GG25], Theorem 3.5)**.** *Let $\mathbb{F}$ be a finite field and $\mathcal{C} \subseteq (\mathbb{F}^s)^n$ be an $\mathbb{F}$-additive code. If $\mathcal{C}$ is $(\delta, a, n+1)$ line-decodable then $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) \le a/|\mathbb{F}|$.*[8]

As it is stronger than MCA, and has already been used to derive new MCA bounds, line decoding may be a useful notion for further understanding of MCA. Additionally, it can be used directly in some IOP constructions in place of MCA (see e.g., [Hab24]), and so considering line decoding rather than MCA, may provide new insights or tighter analysis of protocols. See also the recent paper by Carmon, Goldberg, Haböck, Lerer, and Lesokhin [CGHLL26, Appendix A] for further discussion of line-decoding, how to use it in analysis of IOPs, as well as stated conjectures.

[8] Better MCA bounds can be derived when the code is list decodable. See [GG25] for more details.

5 Connections between list decoding and correlated agreement

In this section we survey known relationships between list decoding and correlated agreement. Finding a formal connection between these two notions is a fundamental question about the nature of correlated agreement. In particular:

- Showing a condition under which list decoding directly implies CA may allow us to shift our focus to improving list decoding bounds.
- Showing that CA implies list decoding would show that proving list decoding bounds is a necessary precondition for proving CA.

In the following we describe what is known about these implications.

List decoding implies CA. Implications of list decoding bounds to CA are implicit in the proof techniques of [BKS18], and an explicit theorem formulating this connection was later shown in [GCXK25], whose results extend to MCA. The following theorem asserts that if a linear code is list decodable at radius δ then CA (in fact, MCA) holds at radius $\approx 1-\sqrt{1-\delta}$ with error parameter related to the list size.

Theorem 5.1 ([GCXK25], Theorem 3)**.** *Let $\mathcal{C} \subseteq \mathbb{F}^n$ be a linear code over a finite field $\mathbb{F}$ and let $\delta, \eta \in (0,1)$ be parameters. If $\mathcal{C}$ has $|\Lambda(\mathcal{C},\delta)| \leq L$ then*

$$\varepsilon_{\mathsf{mca}}\left(\mathcal{C}, 1-\sqrt{1-\delta+\eta}\right) \leq \frac{L^2\delta n + 1/\eta}{|\mathbb{F}|} = O_\delta\left(\frac{L^2 \cdot n}{\eta \cdot |\mathbb{F}|}\right) .$$

When seen through the lens of Reed–Solomon codes, the above theorem implies MCA up to the "2 Johnson bounds" regime for all codes (by applying Corollary 3.3), and, since *random* Reed–Solomon codes are list decodable up to capacity (Theorem 3.6), implies MCA for random Reed–Solomon codes up to the Johnson bound.

Strengthening Theorem 5.1 to remove the square-root loss in proximity would reestablish all known results (up to potentially differences in the error bound). However, as we will see later on in this section, this cannot hold in general as there exist codes that are list decodable but do not exhibit CA at the same radius. The open question must therefore be rephrased to ask whether such implications can be shown to hold for interesting special cases.

CA implies list decoding. The following theorems show that in order to show CA with very small error, it is necessary to show list decoding bounds of roughly the field size. Stated differently, once the list size becomes large compared to the field size, the CA error cannot be very small.

We first present a result which states that showing CA with error $1/2n$, even with a large proximity loss, implies list decoding of size $|\mathbb{F}|$ at a similar proximity.

Theorem 5.2 ([BCHKS25], Theorem 1.9)**.** *Let $\mathcal{C} := \mathsf{RS}[\mathbb{F},\mathcal{L},k]$ be a Reed–Solomon code with rate ρ and let $\delta \in (0, 1-\rho)$ be a proximity parameter. If*

$$\varepsilon_{\mathsf{ca}}\left(\mathcal{C}, \delta_{\mathsf{fld}} = \delta + \frac{2}{n}, \delta_{\mathsf{int}} = 1-\rho-\frac{1}{n}\right) < \frac{1}{2n} ,$$

then $|\Lambda(\mathcal{C},\delta)| < |\mathbb{F}|$.

The theorem below shows that if a Reed–Solomon code has CA error roughly $1/k$ at proximity δ then a related code has list decoding of size $|\mathbb{F}|$ at the same radius.

Theorem 5.3 ([CS25],Theorem 2)**.** *Let* $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$ *and* $\mathcal{C}^+ := \mathsf{RS}[\mathbb{F}, \mathcal{L}, k+1]$ *be Reed–Solomon codes with* $|\mathcal{L}| = n$*. For* $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C}))$ *and* $\eta \in [0, 1)$*, if* $\varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta) \leq \eta \cdot \left(\frac{1}{k} - \frac{n}{k|\mathbb{F}|}\right)$*, then*

$$\left|\Lambda\left(\mathcal{C}^+, \delta\right)\right| \leq \left\lceil \frac{|\mathbb{F}|}{1-\eta} \cdot \varepsilon_{\mathsf{ca}}(\mathcal{C}, \delta) \right\rceil .$$

Limitations. Finally, we present evidence that, in some regimes, correlated agreement and list decoding are separate notions:

Theorem 5.4 ([BGKS20], Lemma 3.3)**.** *For all* $\mathbb{F}$ *of characteristic* 2 *the code* $\mathcal{C} := \mathsf{RS}[\mathbb{F}, \mathbb{F}, |\mathbb{F}|/8]$ *with rate* $\rho = 1/8$ *has*

$$\varepsilon_{\mathsf{ca}}\left(\mathcal{C}, 1 - \rho^{1/3}\right) \geq 1 - \frac{1}{|\mathbb{F}|} .$$

(In fact, the above holds also with proximity loss for $\delta_{\mathsf{fld}} = 1 - \rho^{1/3}$ *and* $\delta_{\mathsf{int}} = 1 - \rho^{2/3}$*.)*

The above theorem shows that there are codes that do not exhibit correlated agreement at $1-\rho^{1/3} = 0.5$. However, by the Johnson bound, at distance 0.55 (i.e., in Theorem 3.2 setting $\eta \approx 0.1$ so that $1 - \sqrt{\rho} - \eta = 0.55$), the code has list size ≈ 40, a constant independent of the field size.

While this example shows that, in general, we cannot expect list decoding to tightly imply correlated agreement, observe that the code is full-domain (i.e., the evaluation domain is the entire field). Thus, one could still hope for such a relationship for codes where the gap between the field size and the evaluation domain size is large (recall that in the SNARK use-case, we usually think of $|\mathbb{F}|$ as being very large compared to n).

6 Toy problem

In this section, we describe a simple toy problem and a protocol for this problem, whose soundness proof captures many of the complexities encountered during the analysis of more complex protocols such as [ACFY25; NA25; BCFW25]. In particular, the analysis of this protocol demonstrates the need for both mutual correlated agreement (MCA) and a list decoding bound. Appendix A contains preliminaries relevant for this section.

6.1 Protocol

The protocol considers a relation consisting of codewords whose underlying messages satisfy a given linear constraint. We refer to this relation to be about *constrained codes.*

Definition 6.1. *Let $\ell \in \mathbb{N}$ and $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a $\mathbb{F}$-additive code. We define the* **toy problem relation** *as:*

$$\mathcal{R}_{\mathcal{C}}^{\ell} \coloneqq \left\{ ((\boldsymbol{v}, \boldsymbol{\mu}), F, \boldsymbol{F}) \;\middle|\; F = \mathcal{C}^{\equiv_\ell}(\boldsymbol{F}) \;\wedge\; \boldsymbol{F} \cdot \boldsymbol{v} = \boldsymbol{\mu} \right\}$$

where $\boldsymbol{v} \in \mathbb{F}^k$, $\boldsymbol{\mu} \in \mathbb{F}^\ell$, $F\colon [n] \to \mathbb{F}^{\ell \cdot s}$, and $\boldsymbol{F} \in \mathbb{F}^{\ell \times k}$.

In the toy protocol, we consider the above relation when $\ell = 2$, in which case $\boldsymbol{\mu} = (\mu_1, \mu_2)$, $F = (f_1, f_2)$, and $\boldsymbol{F} = (\boldsymbol{f}_1, \boldsymbol{f}_2)$. The verifier in the protocol has $(\boldsymbol{v}, \mu_1, \mu_2)$ and oracle access to the (purported) codewords f_1, f_2. It aims to test whether f_1, f_2 belong to the constrained code defined by $(\boldsymbol{v}, \mu_1, \mu_2)$ and the relation $\mathcal{R}_{\mathcal{C}}^2$. To do so, the verifier requests from the prover a random linear combination of the messages $\boldsymbol{f}_1, \boldsymbol{f}_2$, encodes the result as a codeword in $\mathcal{C}$, and checks at a few uniformly random positions that this encoding agrees with the inputs f_1, f_2. The following construction describes this protocol in detail.

Construction 6.2. Let $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a $\mathbb{F}$-additive code and $t \in \mathbb{N}$. We let $\mathcal{T}[\mathcal{C}, t] = (\mathbf{P}, \mathbf{V})$ be the following protocol.

- **Inputs.** The verifier $\mathbf{V}$ receives an explicit input $\mathbb{x} = (\boldsymbol{v}, \mu_1, \mu_2)$ and an implicit input $\mathbb{y} = f_1, f_2\colon [n] \to \mathbb{F}^s$. The honest prover $\mathbf{P}$ receives additionally $\mathbb{w} = (\boldsymbol{f}_1, \boldsymbol{f}_2) \in (\mathbb{F}^k)^2$ such that $\langle \boldsymbol{f}_1, \boldsymbol{v} \rangle = \mu_1$, $\langle \boldsymbol{f}_2, \boldsymbol{v} \rangle = \mu_2$, $f_1 = \mathcal{C}(\boldsymbol{f}_1)$ and $f_2 = \mathcal{C}(\boldsymbol{f}_2)$.
- **Interactive phase.**
  1. **Combination randomness.** The verifier samples and sends $\gamma \leftarrow \mathbb{F}$.
  2. The prover sends $\boldsymbol{g} \in \mathbb{F}^k$. In the honest case, $\boldsymbol{g} \coloneqq \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2$.
  3. **Spot-check randomness.** The verifier samples and sends $x_1, \ldots, x_t \leftarrow [n]$.
- **Decision phase.** The verifier computes $g \coloneqq \mathcal{C}(\boldsymbol{g})$ and checks that:
  1. $\langle \boldsymbol{g}, \boldsymbol{v} \rangle = \mu_1 + \gamma \cdot \mu_2$.
  2. for $i \in [t]$ it holds that $g(x_i) = f_1(x_i) + \gamma \cdot f_2(x_i)$

Argument size. Construction 6.2 can be compiled into a (non-interactive) argument of knowledge via standard techniques [BCS16]. Letting $|\mathsf{H}|$ denote the digest size (in bits) of the Merkle hash function used, the resulting argument size[9] is

$$2t \cdot (|\mathsf{H}| \cdot \log n + s \cdot \log |\mathbb{F}|) \text{ bits} \tag{1}$$

[9] This ignores the size of $\boldsymbol{g}$ which would contribute an additional (dominating) factor of $k \cdot \log |\mathbb{F}|$ bits.

stemming from the Merkle proofs for evaluations of f_1, f_2 in step (2) of the decision phase. To achieve λ bits of security the digest size must satisfy $|\mathsf{H}| \geq 2\lambda$. The above estimates does not include optimizations such as Merkle path pruning[10].

6.2 Soundness

We prove that the toy protocol is (knowledge) sound. Because the verifier only sees a few elements in f_1, f_2, we can only prove soundness with respect to a *relaxed* relation, which we define next.

Definition 6.3. *Let $\ell \in \mathbb{N}$, $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a $\mathbb{F}$-additive code and $\delta \in (0,1)$. We define the* **relaxed toy problem relation** *as:*

$$\tilde{\mathcal{R}}^{\ell}_{\mathcal{C},\delta} := \left\{ (\mathbb{x}, \mathbb{y}, (\mathbb{w}, \mathbb{y}^*)) \;\middle|\; \begin{array}{l} \Delta(\mathbb{y}, \mathbb{y}^*) \leq \delta \\ \wedge\ (\mathbb{x}, \mathbb{y}^*, \mathbb{w}) \in \mathcal{R}^{\ell}_{\mathcal{C}} \end{array} \right\} .$$

The relaxed relation only provides a *proximity guarantee* that the oracle $\mathbb{y}$ is δ-close to a valid instance of the relation. This is both necessary (as the verifier does not read $\mathbb{y}$ in full) and sufficient for applications. We recall the notion of *erasure correction* for a code which will enable us to achieve knowledge soundness in Construction 6.2 for *any linear code* $\mathcal{C}$.

Definition 6.4. *A code $\mathcal{C} \subseteq \Sigma^n$ supports* **erasure correction with correction time** $\mathsf{ecor}_{\mathcal{C}}$ *if there exists a deterministic algorithm $\mathbf{E}_{\mathcal{C}}$ (the* **erasure corrector***) that, given input $f\colon [n] \to \Sigma \cup \{\perp\}$:*

- *if $|f^{-1}(\perp)| < \delta_{\mathsf{min}}(\mathcal{C}) \cdot n$ and there exists a codeword $u \in \mathcal{C}$ such that $f(i) = u(i)$ for all $i \in [n] \setminus f^{-1}(\perp)$, then $\mathbf{E}_{\mathcal{C}}(f) = u$ (note that such a codeword u is unique); and*
- *otherwise, $\mathbf{E}_{\mathcal{C}}(f) = \perp$.*

Moreover, $\mathbf{E}_{\mathcal{C}}$ performs at most $\mathsf{ecor}_{\mathcal{C}}$ field operations.

Every additive code can be erasure decoded in polynomial time.

Lemma 6.5 ([GRS12]). *Every additive code $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ supports erasure correction with correction time $\mathsf{ecor}_{\mathcal{C}} = O((s \cdot n)^3)$*

Lemma 6.6. *For any $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C}))$, the protocol $\mathcal{T}[\mathcal{C}, t]$ (Construction 6.2) has knowledge soundness (Definition A.3) with respect to $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ with extraction time $O(\mathsf{enc}_{\mathcal{C}} + \mathsf{ecor}_{\mathcal{C}})$ and error*

$$\max\left\{ \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda\left(\mathcal{C}^{\equiv_2}, \delta\right)|}{|\mathbb{F}|} ,\ (1-\delta)^t \right\} .$$

The quantity $|\Lambda\left(\mathcal{C}^{\equiv_2}, \delta\right)|$ in the lemma statement is clearly at most $|\Lambda\left(\mathcal{C}, \delta\right)|^2$, and in some cases may be more tightly related to $|\Lambda\left(\mathcal{C}, \delta\right)|$ via Lemma 2.10.

Proof. To prove knowledge soundness of Construction 6.2 we exhibit an extractor that extracts a valid witness from a successful prover. We define the extractor $\mathbf{E}$ as follows:

$\mathbf{E}(\mathbb{y}, (\gamma, \boldsymbol{g}))$:

1. Parse $\mathbb{y} = (f_1, f_2)$ and compute $g := \mathcal{C}(\boldsymbol{g})$.
2. Let $S \subseteq [n]$ be the largest S such that $\Delta_S(f_1 + \gamma \cdot f_2,\ g) = 0$. This S is unique and can be constructed efficiently by a simple greedy strategy.

[10] A crude estimate for the argument size in that setting would be $\approx t \cdot (|\mathsf{H}| \cdot (\log n - \log t) + s \cdot \log |\mathbb{F}|)$.

3. Erasure decode f_1, f_2 with respect to S to obtain messages $\boldsymbol{f_1}, \boldsymbol{f_2} \in \mathbb{F}^k$ such that

$$\Delta_S(f_1, \mathcal{C}(\boldsymbol{f_1})) = \Delta_S(f_2, \mathcal{C}(\boldsymbol{f_2})) = 0.$$

If erasure decoding fails[11] on either f_1 or f_2 then output $\perp$ and abort.

4. Output $(\boldsymbol{f}_1, \boldsymbol{f}_2)$.

We show[12] that for every $\delta \in (0,1)$ and $\mathbb{x}, \mathbb{y}, \tilde{\mathbf{P}}$,

$$\Pr\left[\begin{array}{l} \mathbf{V}^{\mathbb{y}}(\mathbb{x}, (\gamma, \boldsymbol{g}, x_1, \ldots, x_t)) = 1 \\ \wedge \left(\begin{array}{c} \langle \boldsymbol{f_1}, \boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2}, \boldsymbol{v}\rangle \neq \mu_2 \vee \\ \Delta\big((f_1, f_2), \mathcal{C}^{\equiv 2}(\boldsymbol{f_1}, \boldsymbol{f_2})\big) > \delta \end{array}\right) \end{array} \middle| \begin{array}{l} \gamma \leftarrow \mathbb{F}, x_1, \ldots, x_t \leftarrow [n] \\ \boldsymbol{g} := \tilde{\mathbf{P}}(\gamma) \\ (\boldsymbol{f}_1, \boldsymbol{f}_2) := \mathbf{E}(\mathbb{y}, (\gamma, \boldsymbol{g})) \end{array}\right]$$
$$\leq \max\left\{\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda\,(\mathcal{C}^{\equiv 2}, \delta)|}{|\mathbb{F}|}\,,\,(1-\delta)^t\right\}. \tag{2}$$

Write $\mathbb{x} = (\boldsymbol{v}, \mu_1, \mu_2)$ and $\mathbb{y} = (f_1, f_2)$. Define the event E (defined over the choice of γ) that there exists a set $S \subseteq [n]$ with $|S| \geq (1-\delta) \cdot n$ such that $\Delta_S(f_1 + \gamma \cdot f_2, \mathcal{C}) = 0$ and $\Delta_S((f_1, f_2), \mathcal{C}^{\equiv 2}) > 0$. By Definition 4.3, $\Pr_\gamma[E] \leq \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$, so let us assume that $\neg E$ holds. We consider two cases:

First, suppose that $\Delta(f_1 + \gamma \cdot f_2, \mathcal{C}(\boldsymbol{g})) > \delta$. In this case the verifier $\mathbf{V}^{\mathbb{y}}(\mathbb{x}, (\gamma, \boldsymbol{g}, x_1, \ldots, x_t))$ accepts with probability at most $(1-\delta)^t$.

Second, suppose that $\Delta(f_1 + \gamma \cdot f_2, \mathcal{C}(\boldsymbol{g})) \leq \delta$. This means that there exists $S \subseteq [n]$ with $|S| \geq (1-\delta) \cdot n$ such that $\Delta_S(f_1 + \gamma \cdot f_2, \mathcal{C}) = 0$, and therefore, as long as $\neg E$ holds, it must be that $\Delta_S((f_1, f_2), \mathcal{C}^{\equiv 2}) = 0$. Then erasure decoding in the extractor $\mathbf{E}$ will succeed so that its output $(\boldsymbol{f}_1, \boldsymbol{f}_2)$ will satisfy $\Delta_S((f_1, f_2), (\mathcal{C}(\boldsymbol{f}_1), \mathcal{C}(\boldsymbol{f_2}))) = 0$, and thus $\Delta\big((f_1, f_2), \mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f_2})\big) \leq \delta$ as required when the verifier accepts. We are left to bound the probability that $\langle \boldsymbol{f}_1, \boldsymbol{v}\rangle \neq \mu_1$ or $\langle \boldsymbol{f}_2, \boldsymbol{v}\rangle \neq \mu_2$ when the verifier accepts. First we argue that $\langle \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2,\ \boldsymbol{v}\rangle = \mu_1 + \gamma \cdot \mu_2$. To see why, observe that

$$\mathcal{C}(\boldsymbol{g})(S) = (f_1 + \gamma \cdot f_2)(S) = f_1(S) + \gamma \cdot f_2(S) = \mathcal{C}(\boldsymbol{f}_1)(S) + \gamma \cdot \mathcal{C}(\boldsymbol{f}_2)(S) = \mathcal{C}(\boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2)(S) \tag{3}$$

and therefore $\boldsymbol{g} = \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2$ because $\delta < \delta_{\mathsf{min}}(\mathcal{C})$ by assumption. Moreover, since the verifier accepts, it must be that $\langle \boldsymbol{g}, \boldsymbol{v}\rangle = \mu_1 + \gamma \cdot \mu_2$. Therefore $\langle \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2,\ \boldsymbol{v}\rangle = \mu_1 + \gamma \cdot \mu_2$, which for any fixed pair $\boldsymbol{f}_1, \boldsymbol{f}_2$ such that $\langle \boldsymbol{f}_1, \boldsymbol{v}\rangle \neq \mu_1$ or $\langle \boldsymbol{f}_2, \boldsymbol{v}\rangle \neq \mu_2$ only occurs with probability at most $\frac{1}{|\mathbb{F}|}$ over the choice of γ by the polynomial identity lemma. Since (assuming $\neg E$ holds) we have that $\Delta\big((f_1, f_2), \mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2)\big) \leq \delta$ it must be that $\mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2) \in \Lambda(\mathcal{C}^{\equiv 2}, \delta, (f_1, f_2))$ and so there are at most $|\Lambda\,(\mathcal{C}^{\equiv 2}, \delta)|$ such choices for $(\boldsymbol{f}_1, \boldsymbol{f}_2)$. Now a union bound gives the error term $|\Lambda\,(\mathcal{C}^{\equiv 2}, \delta)|\,/|\mathbb{F}|$, which concludes the proof that (2) holds.

To make this argument formal, let us write for convenience

$$P := \Pr\left[\Delta(f_1 + \gamma \cdot f_2, \mathcal{C}(\boldsymbol{g})) > \delta \middle| \begin{array}{l} \gamma \leftarrow \mathbb{F} \\ \boldsymbol{g} := \tilde{\mathbf{P}}(\gamma) \\ (\boldsymbol{f}_1, \boldsymbol{f}_2) := \mathbf{E}(\mathbb{y}, (\gamma, \boldsymbol{g})) \end{array}\right].$$

[11] Failure can happen, for example, if S is too small, that is $\left|S\right| < (1 - \delta_{\mathsf{min}}(\mathcal{C})) \cdot n$.

[12] Note that this is a stronger notion than what Definition A.3 requires, as the extractor is not allowed to read $\mathbb{x}$ nor the final transcript, and does not output $\mathbb{y}^*$. The final extractor is obtained by letting $\mathbf{E}'(\mathbb{x}, \mathbb{y}, \mathsf{tr})$ be the algorithm that parses $\mathsf{tr} = (\gamma, \boldsymbol{g}, (x_1, \ldots, x_t))$, compute $\mathbb{w} = \mathbf{E}(\mathbb{y}, (\gamma, \boldsymbol{g}))$ and outputs $(\mathbb{w}, \mathcal{C}^{\equiv 2}(\mathbb{w}))$. $\mathbf{E}'$ performs $O(\mathsf{enc}_{\mathcal{C}} + \mathsf{t}_{\mathbf{E}})$ operations.

Then putting it all together we obtain that:

$$
\begin{aligned}
&\Pr\left[\begin{array}{l} \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g}))\end{array}\right] \\
&= \Pr\left[\begin{array}{l} \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))>\delta \\ \wedge\ \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g}))\end{array}\right] \\
&+ \Pr\left[\begin{array}{l} \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))\leq\delta \\ \wedge\ \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g}))\end{array}\right] \\
&= \Pr\left[\begin{array}{l} \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g})) \\ \hline \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))>\delta\end{array}\right]\cdot P \\
&+ \Pr\left[\begin{array}{l} \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g})) \\ \hline \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))\leq\delta\end{array}\right]\cdot (1-P) \\
&\leq \max\left\{(1-\delta)^t\,,\ \Pr\left[\begin{array}{l} \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g})) \\ \hline \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))\leq\delta\end{array}\right]\right\} \\
&\leq \max\left\{(1-\delta)^t\,,\ \Pr[E]+\Pr\left[\begin{array}{l} \neg E \\ \wedge\ \mathbf{V}^{\mathbb{y}}(\mathbb{x},(\gamma,\boldsymbol{g},x_1,\ldots,x_t))=1 \\ \wedge\begin{pmatrix}\langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2 \\ \Delta((f_1,f_2),\mathcal{C}^{\equiv_2}(\boldsymbol{f}_1,\boldsymbol{f_2}))>\delta\end{pmatrix}\end{array}\middle|\begin{array}{l}\gamma\leftarrow\mathbb{F},x_1,\ldots,x_t\leftarrow[n]\\ \boldsymbol{g}\coloneqq\tilde{\mathbf{P}}(\gamma)\\ (\boldsymbol{f}_1,\boldsymbol{f}_2)\coloneqq\mathbf{E}(\mathbb{y},(\gamma,\boldsymbol{g})) \\ \hline \Delta(f_1+\gamma\cdot f_2,\mathcal{C}(\boldsymbol{g}))\leq\delta\end{array}\right]\right\} \\
&\leq \max\left\{(1-\delta)^t\,,\ \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)+\Pr\left[\begin{array}{l} \neg E \\ \wedge\ \exists(\boldsymbol{f}_1,\boldsymbol{f}_2)\text{ s.t. }(\mathcal{C}(\boldsymbol{f}_1),\mathcal{C}(\boldsymbol{f}_2))\in\Lambda(\mathcal{C}^{\equiv_2},\delta,(f_1,f_2)) \\ \wedge\ \Big(\ \langle \boldsymbol{f_1},\boldsymbol{v}\rangle \neq \mu_1 \vee \langle \boldsymbol{f_2},\boldsymbol{v}\rangle\neq\mu_2\ \Big) \\ \wedge\ \langle \boldsymbol{f_1}+\gamma\cdot\boldsymbol{f}_2,\boldsymbol{v}\rangle=\mu_1+\gamma\cdot\mu_2 \end{array}\right]\right\} \\
&\leq \max\left\{(1-\delta)^t\,,\ \varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)+\frac{|\Lambda\left(\mathcal{C}^{\equiv_2},\delta\right)|}{|\mathbb{F}|}\right\}\,,
\end{aligned}
$$

where we have used that for $\lambda\in[0,1]$ it holds that $a\cdot\lambda+b\cdot(1-\lambda)\leq\max\{a,b\}$. □

Remark 6.7. The soundness argument we used to prove Lemma 6.6 relies heavily on mutual correlated agreement (MCA). The weaker notion of correlated agreement (CA) is insufficient to prove the lemma, even if we replace erasure decoding in the extractor with list decoding. The problem is that there could exist an (f_1,f_2) that is δ-*close* to $\mathcal{C}^{\equiv_2}$, but no $(u_1,u_2)\in\Lambda(\mathcal{C}^{\equiv_2},\delta,(f_1,f_2))$

satisfies the constraints $\langle \mathcal{C}^{-1}(u_1), \boldsymbol{v}\rangle = \mu_1$ and $\langle \mathcal{C}^{-1}(u_2), \boldsymbol{v}\rangle = \mu_2$. This would violate the soundness of the protocol. What breaks in the proof of Lemma 6.6 is that when using CA, instead of MCA, we can no longer prove that every codeword $u \in \Lambda(\mathcal{C}, f_1 + \gamma \cdot f_2, \delta)$ can be written as $u = u_1 + \gamma \cdot u_2$ for codewords $(u_1, u_2) \in \Lambda(\mathcal{C}^{\equiv 2}, (f_1, f_2), \delta)$. Mutual correlated agreement (MCA) lets us prove this fact quite easily, as we do in (3), except with probability $\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$.

Lemma 6.8. *For any $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C}))$, the protocol $\mathcal{T}[\mathcal{C}, t]$ (Construction 6.2) has round-by-round knowledge soundness (Definition A.5) with respect to $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ with total extraction time $O(\mathsf{enc}_{\mathcal{C}} + \mathsf{ecor}_{\mathcal{C}})$ and errors*

$$\left(\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda\left(\mathcal{C}^{\equiv 2}, \delta\right)|}{|\mathbb{F}|}, \quad (1-\delta)^t\right)$$

Proof. We define the state function and bound the transition probability next.

- **Initial input.** We set $\mathsf{State}(\mathbb{x}, \mathbb{y}, \emptyset, \mathsf{w}) = 1$ if and only if $(\mathbb{x}, \mathbb{y}, \mathbb{w}) \in \tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$. That is, $\mathsf{State}(\mathbb{x}, \mathbb{y}, \emptyset, \mathsf{w}) = 1$ if and only if $\mathsf{w} = (\boldsymbol{f}_1, \boldsymbol{f}_2, \overline{f}_1, \overline{f}_2)$ such that $\Delta((f_1, f_2), (\overline{f}_1, \overline{f}_2)) \leq \delta$, $(\overline{f}_1, \overline{f}_2) = \mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2)$, $\langle \boldsymbol{f}_1, \boldsymbol{v}\rangle = \mu_1$, and $\langle \boldsymbol{f}_2, \boldsymbol{v}\rangle = \mu_2$.

- **Combination randomness.** At this stage, the transcript tr is empty and the verifier samples $\gamma \leftarrow \mathbb{F}$. We set $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w}) = 1$ if and only if $\mathsf{w} = \boldsymbol{f}$ such that

  – $\langle \boldsymbol{f}, \boldsymbol{v}\rangle = \mu_1 + \gamma \cdot \mu_2$ and
  – $\Delta(f_1 + \gamma \cdot f_2, \mathcal{C}(\boldsymbol{f})) \leq \delta$.

  We define the extractor $\mathbf{E}$ as follows:

  $\mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w})$:
  1. Parse $\mathsf{w} = \boldsymbol{f}$ and compute $f = \mathcal{C}(\boldsymbol{f})$.
  2. Let $S \subseteq [n]$ be the largest S such that $f(S) = f_1(S) + \gamma \cdot f_2(S)$.
  3. Erasure decode f_1, f_2 with respect to S to obtain $\boldsymbol{f}_1, \boldsymbol{f}_2$.
  4. Output $(\boldsymbol{f}_1, \boldsymbol{f}_2, \mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2))$.

  We show that

  $$\Pr_{\gamma \leftarrow \mathbb{F}}\left[\exists \mathsf{w} : \begin{array}{l} \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w})) = 0 \\ \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w}) = 1 \end{array}\right] \leq \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda\left(\mathcal{C}^{\equiv 2}, \delta\right)|}{|\mathbb{F}|}.$$

  Define the event E (defined over the choice of γ) that there exists a set $S \subseteq [n]$ with $|S| \geq (1-\delta) \cdot n$ such that $\Delta_S(f_1 + \gamma \cdot f_2, \mathcal{C}) = 0$ and $\Delta_S((f_1, f_2), \mathcal{C}^{\equiv 2}) > 0$. By Definition 4.3, $\Pr_\gamma[E] \leq \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$.

  Assuming that $\neg E$ holds, fix a $\mathsf{w} = \boldsymbol{f}$ such that $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w}) = 1$.

  Since $\Delta(f_1 + \gamma \cdot f_2, \mathcal{C}(\boldsymbol{f})) \leq \delta$, there exists a set S of size $|S| \geq (1-\delta) \cdot n$ such that $\Delta_S(f_1 + \gamma \cdot f_2, \mathcal{C}) = 0$. Take S to be maximal among such sets. Then, by virtue of $\neg E$ holding, it must be that $\Delta_S((f_1, f_2), \mathcal{C}^{\equiv 2}) = 0$, and so there exists $(\boldsymbol{f}_1, \boldsymbol{f}_2)$ such that $(f_1, f_2)(S) = \mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2)(S)$.

  Note in particular that it must be that $(\boldsymbol{f}_1, \boldsymbol{f}_2)$ are those output by $\mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\gamma, \mathsf{w})$ (as by the definition of the extractor S is the same set as in the analysis, and the codewords output agree on a large set). This implies that there are at most $|\Lambda\left(\mathcal{C}^{\equiv 2}, \delta\right)|$ possible pairs of $(\boldsymbol{f}_1, \boldsymbol{f}_2)$ that $\mathbf{E}$ outputs (as we have seen that $\mathcal{C}^{\equiv 2}(\boldsymbol{f}_1, \boldsymbol{f}_2) \in \Lambda(\mathcal{C}^{\equiv 2}, \delta, (f_1, f_2))$.)

  Further, it must be that $\boldsymbol{f} = \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2$. This is because by definition we have $\mathcal{C}(\boldsymbol{f})(S) = f_1(S) + \gamma \cdot f_2(S) = \mathcal{C}(\boldsymbol{f}_1)(S) + \gamma \cdot \mathcal{C}(\boldsymbol{f}_2)(S) = \mathcal{C}(\boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2)(S)$ and since $|S| \geq (1-\delta) \cdot n \geq (1 - \delta_{\mathsf{min}}(\mathcal{C})) \cdot n$ it must be that the right and left hand side are the same codeword.

Next, we show that with high probability it must be that $\langle \boldsymbol{f}_1, \boldsymbol{v} \rangle = \mu_1$ and $\langle \boldsymbol{f}_2, \boldsymbol{v} \rangle = \mu_2$. If not, by the polynomial identity lemma unless with probability at most $\frac{1}{|\mathbb{F}|}$ it holds that $\langle \boldsymbol{f}, \boldsymbol{v} \rangle = \langle \boldsymbol{f}_1, \boldsymbol{v} \rangle + \gamma \cdot \langle \boldsymbol{f}_2, \boldsymbol{v} \rangle \neq \mu_1 + \gamma \cdot \mu_2$, contradicting $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| \gamma, \mathsf{w}) = 1$. Since there are at most $|\Lambda(\mathcal{C}^{\equiv_2}, \delta)|$ such pairs, this concludes the analysis.

- **Spot-check randomness.** At this stage, the transcript $\mathsf{tr} = (\gamma, \boldsymbol{g})$ and the verifier samples $x_1, \ldots, x_t \leftarrow [n]$. We set $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \mathsf{w}) = 1$ if and only if $\mathsf{w} = \boldsymbol{f}$ such that

  – $\langle \boldsymbol{f}, \boldsymbol{v} \rangle = \mu_1 + \gamma \cdot \mu_2$.
  – for $i \in [t]$, $\mathcal{C}(\boldsymbol{f})(x_i) = f_1(x_i) + \gamma \cdot f_2(x_i)$.

  We define the extractor $\mathbf{E}$ to be the extractor $\mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \emptyset) = \boldsymbol{g}$.

  We show that

$$\Pr_{x_1,\ldots,x_t \leftarrow [n]} \left[\exists \mathsf{w} : \begin{array}{l} \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \mathsf{w})) = 0 \\ \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \mathsf{w}) = 1 \end{array} \right] \leq (1 - \delta)^t .$$

  If $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \boldsymbol{g}) = 0$, then either:

  1. $\langle \boldsymbol{g}, \boldsymbol{v} \rangle \neq \mu_1 + \gamma \cdot \mu_2$; or
  2. $\Delta(g, f_1 + \gamma \cdot f_2) > \delta$ where $g \coloneqq \mathcal{C}(\boldsymbol{g})$.

  If the first case holds, $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \emptyset) = 0$, and we are done. In the second case, except with probability at most $(1 - \delta)^t$, for some $j \in [t]$ it holds that $g(x_j) \neq f_1(x_j) + \gamma \cdot f_2(x_j)$, and so again $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr} \| (x_1, \ldots, x_t), \emptyset) = 0$.

$\square$

6.3 Concrete parametrizations

We consider the general setting where

- $\mathbb{F}$ is a sextic extension of $\mathbb{B} = \mathbb{F}_q$ where $q = 2^{31} - 2^{24} + 1$ (the Koala Bear prime);
- $k = 2^{20}$, $s \cdot n = 2^{21}$ so that $\rho = 1/2$ and;
- $t = 128$.

We analyze the protocol with two different families of codes. In both cases, the code $\mathcal{C}$ is an extension code of a $\mathcal{C}_{\mathbb{B}}$ defined over the base field $\mathbb{B}$. In the common case where input messages are defined over the base field, and are committed within the same Merkle tree, (1) reduces to

$$128 \cdot (256 \cdot \log n + 2 \cdot 31 \cdot s) \text{ bits.}$$

The above expression is minimized when $n = 2^{18}$ and $s = 2^3$ in which case the result is 79.75 KiB.

We analyze the knowledge soundness error and argument size of Construction 6.2 under different choices of code $\mathcal{C}$ (and consequently of n and s).

6.3.1 Interleaved Reed–Solomon codes

The first instantiation is via the interleaving of Reed–Solomon codes.

- $\mathcal{L} \subseteq \mathbb{B}$ is a smooth domain with $|\mathcal{L}| = n$.
- $\mathcal{C} \coloneqq \mathsf{IRS}[\mathbb{F}, \mathcal{L}, k, s]$ and $\mathcal{C}_{\mathbb{B}} \coloneqq \mathsf{IRS}[\mathbb{B}, \mathcal{L}, k, s]$.

This instantiation closely resembles concretely deployed instances of hash-based proof systems.

Let $\eta > 0$ and set $\delta := 1-\sqrt{\rho}-\eta$ and $m := \max\left\{\left\lceil \frac{\sqrt{\rho}}{2\eta} \right\rceil, 3\right\}+1/2$. By Theorem 4.12 and lemma 4.7 we have that

$$\begin{aligned}
\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) &\le s \cdot \varepsilon_{\mathsf{mca}}(\mathsf{RS}[\mathbb{F}, \mathcal{L}, k/s], \delta) \\
&\le \frac{2 \cdot m^5 + 3 \cdot m \cdot \delta \cdot \rho}{3 \cdot \rho^{3/2} \cdot |\mathbb{F}|} \cdot s \cdot n + \frac{s \cdot m}{\sqrt{\rho} \cdot |\mathbb{F}|} \\
&\le \frac{2 \cdot m^5 + 3 \cdot m \cdot \delta \cdot \rho}{2^{165}} + \frac{s \cdot m}{2^{185.5}},
\end{aligned}$$

and by Theorem 3.2 that:

$$|\Lambda\left(\mathcal{C}^{\equiv_2}, \delta\right)| \le \frac{1}{2 \cdot \eta \cdot \sqrt{\rho}} = \frac{\sqrt{2}}{2 \cdot \eta}.$$

Applying this to Lemma 6.8, we obtain that the round by round knowledge errors of Construction 6.2 are

$$\begin{aligned}
&\min_{\delta \in (0,1)} \left(\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda\left(\mathcal{C}^{\equiv_2}, \delta\right)|}{|\mathbb{F}|},\ (1-\delta)^t \right) \\
&\le \min_{\eta \in (0,1)} \left(\frac{2 \cdot m^5 + 3 \cdot m \cdot \delta \cdot \rho}{2^{165}} + \frac{s \cdot m}{2^{185.5}} + \frac{2^{-186.5}}{\eta},\ (1/\sqrt{2} + \eta)^t \right).
\end{aligned}$$

Fixing $t = 128$. First note that for any $\eta > 0$ it holds that $(1/\sqrt{2}+\eta)^t > (1/\sqrt{2})^{128} = 2^{-64}$, so this analysis can offer *at most* 64 bits of security. For our parameters, the value of η which minimizes the above is $\eta = 1/n$, as the second term dominates the soundness error.

Taking $s = 2^0$ as an example, then $m \approx 2^{18.5}$ and the soundness error is upper bounded by

$$\left(\frac{2^{93.5} + 2^{18.3}}{2^{165}} + \frac{2^{18.5}}{2^{185.5}} + \frac{2^{-186.5}}{2^{-21}},\ (1/\sqrt{2} + 2^{-21})^t \right) \approx (2^{-71.5}, 2^{-64.00}).$$

So, the protocol provably provides approximately 64 bits of security in this setting.

For different choices of s we obtain Table 2.

Argument size when enforcing 128-bits of security. Conversely, we analyze the argument size Construction 6.2 provides when enforcing 128-bits of provable knowledge soundness. Let $s \in \mathbb{N}$ be a folding parameter. We are tasked to find the settings of $t \in \mathbb{N}$ and $\eta > 0$ that minimizes the argument size, that is:

$$t \cdot (256 \cdot \log n + 62 \cdot s) \text{ subject to } \begin{cases} \frac{2 \cdot m^5 + 3 \cdot m \cdot \delta \cdot \rho}{2^{165}} + \frac{s \cdot m}{2^{185.5}} + \frac{2^{-186.5}}{\eta} & \le 2^{-128} \\ (1/\sqrt{2} + \eta)^t & \le 2^{-128} \end{cases}$$

For example, when $s = 2^0$ the minimizing pair is $(\eta, t) \approx (2^{-8.57}, 259)$. In this case, the argument size 171.9 KiB and we have that $m \approx 2^{7.08}$, so the knowledge soundness error is upper bounded by

$$\left(\frac{2^{36.5} + 2^{5.9}}{2^{165}} + \frac{2^{7.08}}{2^{185.5}} + \frac{2^{-186.5}}{2^{-8.57}},\ (1/\sqrt{2} + 2^{-8.57})^{259} \right) \approx (2^{-128.59}, 2^{-128.11}).$$

We detail the result for different values of s in Table 3.

s	η	bits	size
2^0	2^{-21}	$2^{-64.00}$	85.0 KiB
2^1	2^{-20}	$2^{-64.00}$	81.9 KiB
2^2	2^{-19}	$2^{-64.00}$	79.9 KiB
2^3	2^{-18}	$2^{-64.00}$	79.8 KiB
2^4	2^{-17}	$2^{-64.00}$	83.5 KiB
2^5	2^{-16}	$2^{-64.00}$	95.0 KiB
2^6	2^{-15}	$2^{-63.99}$	122.0 KiB
2^7	2^{-14}	$2^{-63.98}$	180.0 KiB
2^8	2^{-13}	$2^{-63.97}$	300.0 KiB
2^9	2^{-12}	$2^{-63.94}$	544.0 KiB
2^{10}	2^{-11}	$2^{-63.87}$	1.01 MiB
2^{11}	2^{-10}	$2^{-63.75}$	1.98 MiB
2^{12}	2^{-9}	$2^{-63.49}$	3.91 MiB

Table 2: Round-by-round knowledge soundness of Construction 6.2 when instantiated with an interleaved Reed–Solomon code $\mathsf{IRS}[\mathbb{F},\mathcal{L},k,s]$. η refers to the value of η which minimizes the soundness error, which in this case is $\eta = 1/n$. The horizontal line marks the point after which the argument size monotonically increases.

6.3.2 Subspace-design codes

The second instantiation is via subspace-design codes. Let $\mathcal{C}\colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a τ-subspace-design code of $\rho = 1/2$. For concreteness, consider the folded Reed–Solomon case where:

- $\mathcal{L} \subseteq \mathbb{B}$ is a smooth domain with $|\mathcal{L}| = n$ and $\omega \in \mathbb{F}$ is $(\mathcal{L}, s)$-admissible.
- $\mathcal{C} := \mathsf{FRS}[\mathbb{F},\mathcal{L},k,s,\omega]$ and $\mathcal{C}_{\mathbb{B}} := \mathsf{FRS}[\mathbb{B},\mathcal{L},k,s,\omega]$.
- τ is defined $\tau(r) = \frac{s\cdot\rho}{s-r+1}$ for $1 \le r \le s$ and $\tau(r) = 1$ otherwise (as in Theorem 2.18).

Fix $r \in \mathbb{N}$ and $\delta := 1 - \tau(r+1) - \frac{3}{2r}$. By Theorem 4.13

$$\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta) \le \frac{r\cdot n + 4r^3}{|\mathbb{F}|},$$

and, since $\delta \le 1 - \tau(r+1) - \frac{1}{r+1}$, by Theorem 3.4 we have

$$|\Lambda(\mathcal{C},\delta)| \le \left|\Lambda\left(\mathcal{C}, 1-\tau(r+1) - \frac{1}{r+1}\right)\right| \le (1-\tau(r+1))\cdot(r+1)\,.$$

To bound the term required by Lemma 6.8, define

$$b_r := \left\lceil \frac{\delta}{\delta_{\mathsf{min}}(\mathcal{C}) - \delta} \right\rceil, \qquad c_r := \left\lceil \log\frac{\delta_{\mathsf{min}}(\mathcal{C})}{\delta_{\mathsf{min}}(\mathcal{C}) - \delta} \right\rceil.$$

Then, by Lemma 2.10,

$$|\Lambda(\mathcal{C}^{\equiv 2},\delta)| \le \binom{b_r + c_r}{c_r}\cdot|\Lambda(\mathcal{C},\delta)|^{c_r} \le \binom{b_r+c_r}{c_r}\cdot((1-\tau(r+1))\cdot(r+1))^{c_r}\,.$$

s	η	t	bits	size
2^0	$2^{-8.57}$	259	$2^{-128.11}$	171.9 KiB
2^1	$2^{-8.57}$	259	$2^{-128.11}$	165.8 KiB
2^2	$2^{-8.57}$	259	$2^{-128.11}$	161.6 KiB
2^3	$2^{-8.57}$	259	$2^{-128.11}$	161.4 KiB
2^4	$2^{-8.57}$	259	$2^{-128.11}$	169.0 KiB
2^5	$2^{-8.56}$	259	$2^{-128.11}$	192.2 KiB
2^6	$2^{-8.56}$	259	$2^{-128.10}$	246.9 KiB
2^7	$2^{-8.60}$	259	$2^{-128.14}$	364.2 KiB
2^8	$2^{-8.65}$	259	$2^{-128.18}$	607.0 KiB
2^9	$2^{-8.43}$	260	$2^{-128.47}$	1.08 MiB
2^{10}	$2^{-8.36}$	260	$2^{-128.39}$	2.06 MiB
2^{11}	$2^{-8.42}$	260	$2^{-128.46}$	4.01 MiB
2^{12}	$2^{-8.55}$	259	$2^{-128.09}$	7.91 MiB

Table 3: Round-by-round knowledge soundness and argument size of Construction 6.2 when instantiated with an interleaved Reed–Solomon code $\mathsf{IRS}[\mathbb{F}, \mathcal{L}, k, s]$. η and t refers to the value which minimizes the soundness error, subject to achieving 128-bits of security. The horizontal line marks the point after which the argument size monotonically increases.

Applying Lemma 6.8, we obtain that the round by round knowledge errors of Construction 6.2 are

$$\begin{aligned}
&\min_{\delta \in (0,1)} \left(\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda(\mathcal{C}^{\equiv_2}, \delta)|}{|\mathbb{F}|},\ (1-\delta)^t \right) \\
&\leq \min_{r \in \mathbb{N}} \left(\frac{r \cdot n + 4r^3}{|\mathbb{F}|} + \frac{\binom{b_r + c_r}{c_r} \cdot ((1 - \tau(r+1)) \cdot (r+1))^{c_r}}{|\mathbb{F}|},\ \left(\tau(r+1) + \frac{3}{2r}\right)^t \right).
\end{aligned}$$

Note also that for $s \leq 2^4$ and $r \in \mathbb{N}$, it holds that $\tau(r+1) + \frac{3}{2r} \geq 1$, and so in those cases no soundness can be proven.

Fixing $t = 128$. Consider $s = 2^5$ as an example, the soundness error is minimized by $r = 8$, in which case $\tau(r+1) = \frac{2^5 \cdot \rho}{2^5 - 9 + 1} = 2/3$. Further, $\delta_{\mathsf{min}}(\mathcal{C}) - \delta \approx 17/48$, so $b_r = 1$ and $c_r = 1$. Then, the round-by-round knowledge soundness errors are upperbounded by

$$\left(\frac{8 \cdot 2^{16} + 4 \cdot 8^3}{2^{186}} + \frac{\binom{1+1}{1} \cdot (1/3 \cdot 9)}{2^{186}},\ (2/3 + 3/16)^{128} \right) \approx (2^{-166.8}, 2^{-29.11}).$$

We optimize the above bound over integers $r \in \mathbb{N}$ for each folding parameter s. We report the minimizing r and the resulting soundness error in Table 4. For small folding parameters ($s \leq 2^4$) the constraint $\tau(r+1) + \frac{3}{2r} < 1$ cannot be met, so we start the table at $s = 2^5$.

Argument size when enforcing 128-bits of security. We are tasked to find the settings of $t \in \mathbb{N}$ and $r \in \mathbb{N}$ that minimizes the argument size, that is:

$$t \cdot (256 \cdot \log n + 62 \cdot s) \text{ subject to } \begin{cases} \frac{r \cdot n + 4r^3}{|\mathbb{F}|} + \frac{\binom{b_r + c_r}{c_r} \cdot ((1-\tau(r+1)) \cdot (r+1))^{c_r}}{|\mathbb{F}|} \leq 2^{-128} \quad \text{and} \\ \left(\tau(r+1) + \frac{3}{2r}\right)^t \leq 2^{-128} \end{cases}$$

s	r	bits	size
2^5	8	$2^{-29.11}$	95.0 KiB
2^6	11	$2^{-55.57}$	122.0 KiB
2^7	17	$2^{-75.39}$	180.0 KiB
2^8	25	$2^{-90.04}$	300.0 KiB
2^9	36	$2^{-100.76}$	544.0 KiB
2^{10}	53	$2^{-108.53}$	1.01 MiB
2^{11}	75	$2^{-114.13}$	1.98 MiB
2^{12}	108	$2^{-118.14}$	3.91 MiB

Table 4: Round-by-round knowledge soundness of Construction 6.2 when instantiated with a folded Reed–Solomon code $\mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega]$. The list size term is bounded via Theorem 3.4 and lemma 2.10. r refers to the value of r which minimizes the soundness error. The horizontal line marks the point after which the soundness error improves over the interleaved Reed–Solomon equivalent.

where b_r, c_r are as defined above.

For example, when $s = 2^5$ the minimizing pair is $(t, r) = (563, 8)$. In this case, the argument size is 417.9 KiB and the round-by-round soundness errors are

$$\left(\frac{8 \cdot 2^{16} + 4 \cdot 8^3}{2^{186}} + \frac{\binom{1+1}{1} \cdot ((1 - 2/3) \cdot 9)}{2^{186}},\ (2/3 + 3/16)^{563}\right) \approx (2^{-168.8}, 2^{-128.03})\,.$$

The resulting sizes are listed in Table 5.

s	r	t	bits	size
2^5	8	563	$2^{-128.03}$	417.9 KiB
2^6	11	295	$2^{-128.07}$	281.2 KiB
2^7	16	218	$2^{-128.22}$	306.6 KiB
2^8	25	182	$2^{-128.02}$	426.6 KiB
2^9	32	163	$2^{-128.01}$	692.8 KiB
2^{10}	50	151	$2^{-128.00}$	1.19 MiB
2^{11}	61	144	$2^{-128.03}$	2.22 MiB
2^{12}	86	139	$2^{-128.01}$	4.25 MiB

Table 5: Round-by-round knowledge soundness of Construction 6.2 when instantiated with a folded Reed–Solomon code $\mathsf{FRS}[\mathbb{F}, \mathcal{L}, k, s, \omega]$. The list size term is bounded via Theorem 3.4 and Lemma 2.10. r, t refers to the value of r and t which minimizes the argument size, while achieving 128-bits of security.

6.4 Attacks

Instead of analyzing the attack on Construction 6.2, we present a simplified construction, which is a *interactive oracle reduction* (IOR) from $\mathcal{R}^2_{\mathcal{C}}$ to $\mathcal{R}^1_{\mathcal{C}}$.

Construction 6.9. Let $\mathcal{C} \colon \mathbb{F}^k \to (\mathbb{F}^s)^n$ be a $\mathbb{F}$-additive code and $t \in \mathbb{N}$.

- **Inputs.** The verifier $\mathbf{V}$ receives as explicit input $\mathbb{x} = (\boldsymbol{v}, \mu_1, \mu_2)$ and as implicit input $\mathbb{y} = f_1, f_2 \colon [n] \to \mathbb{F}^s$. The honest prover $\mathbf{P}$ receives additionally $\mathbb{w} = (\boldsymbol{f}_1, \boldsymbol{f}_2) \in (\mathbb{F}^k)^2$ such that $\langle \boldsymbol{f}_1, \boldsymbol{v} \rangle = \mu_1$, $\langle \boldsymbol{f}_2, \boldsymbol{v} \rangle = \mu_2$, $f_1 = \mathcal{C}(\boldsymbol{f}_1)$, and $f_2 = \mathcal{C}(\boldsymbol{f}_2)$.
- **Interactive phase.**
  1. **Combination randomness.** The verifier samples and sends $\gamma \leftarrow \mathbb{F}$.
- **New instance.** The verifier sets and outputs $\mathbb{x}^* := (\boldsymbol{v}, \mu_1 + \gamma \cdot \mu_2)$, $\mathbb{y}^* := f_1 + \gamma \cdot f_2$. The honest prover sets $\mathbb{w}^* := \boldsymbol{f}_1 + \gamma \cdot \boldsymbol{f}_2$.

It is easy to see by adapting Lemma 6.8 that Construction 6.9 has the following soundness error.

Lemma 6.10. *For any $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C}))$, the protocol $\mathcal{T}[\mathcal{C}, t]$ (Construction 6.9) has knowledge soundness (Definition A.5) from $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ to $\tilde{\mathcal{R}}^1_{\mathcal{C},\delta}$ with total extraction time $O(\mathsf{enc}_{\mathcal{C}} + \mathsf{ecor}_{\mathcal{C}})$ and error*

$$\varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta) + \frac{|\Lambda(\mathcal{C}^{\equiv_2}, \delta)|}{|\mathbb{F}|} .$$

We define the winning set of Construction 6.9 to be the set of challenges for which the prover can successfully reduce from an invalid instance to a valid instance.

Definition 6.11. *Let:*

$$\Omega^{f_1,f_2}_{\boldsymbol{v},\mu_1,\mu_2} = \left\{ \gamma \in \mathbb{F} \,\middle|\, (f_1 + \gamma \cdot f_2) \in \tilde{\mathcal{R}}^1_{\mathcal{C},\delta}[\boldsymbol{v}, \mu_1 + \gamma\mu_2] \right\} .$$

Clearly, the soundness error of Construction 6.9 from $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ to $\tilde{\mathcal{R}}^1_{\mathcal{C},\delta}$ is exactly

$$\max_{\mathbb{x},\mathbb{y}} \frac{|\Omega^{\mathbb{y}}_{\mathbb{x}}|}{|\mathbb{F}|} \text{ subject to } \Delta(\mathbb{y}, \mathcal{R}^2_{\mathcal{C}}[\mathbb{x}]) > \delta .$$

6.4.1 List decoding lower bound

We show that the existence of a word with a large list of close codewords directly leads to an attack on the soundness of Construction 6.2[13]. The attack exploits the fact that, once the verifier has fixed the random linear combination it wants to test, the adversary may select whichever nearby codeword in the list is most convenient.

Lemma 6.12. *Let $\delta \in (0,1)$ be such that $|\mathbb{F}| > \binom{|\Lambda(\mathcal{C}^{\equiv_2},\delta)|}{2}$. Then, the soundness error of Construction 6.9 from $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ to $\tilde{\mathcal{R}}^1_{\mathcal{C},\delta}$ is at least*

$$\frac{|\Lambda(\mathcal{C}^{\equiv_2}, \delta)|}{|\mathbb{F}| + |\Lambda(\mathcal{C}^{\equiv_2}, \delta)| - 1} .$$

Proof. Fix $\delta \in (0,1)$ and choose f_1, f_2 such that $|\Lambda(\mathcal{C}^{\equiv_2}, \delta, (f_1, f_2))| = |\Lambda(\mathcal{C}^{\equiv_2}, \delta)|$. Define the set S and, for $\boldsymbol{v} \in \mathbb{F}^k$, the map $\phi_{\boldsymbol{v}} \colon S \to \mathbb{F}^2$ and the set $S_{\boldsymbol{v}} \subseteq \mathbb{F}^2$ as follows:

$$S := \{(\boldsymbol{f}_1, \boldsymbol{f_2}) \mid \mathcal{C}^{\equiv_2}(\boldsymbol{f}_1, \boldsymbol{f_2}) \in \Lambda(\mathcal{C}^{\equiv_2}, \delta, (f_1, f_2))\} ,$$
$$\phi_{\boldsymbol{v}}(\boldsymbol{f}_1, \boldsymbol{f_2}) := (\langle \boldsymbol{f}_1, \boldsymbol{v} \rangle, \langle \boldsymbol{f}_2, \boldsymbol{v} \rangle) , \; S_{\boldsymbol{v}} := \phi_{\boldsymbol{v}}(S) .$$

[13] A similar observation (applied to codes over extension fields) previously appeared in [FS25]

Note that $|S| = |\Lambda(\mathcal{C}^{\equiv_2}, \delta)|$ and clearly for any $\boldsymbol{v}$ it holds that $|S_{\boldsymbol{v}}| \le |S|$. Further, for any distinct $\boldsymbol{F}_1, \boldsymbol{F}_2 \in S$ we have that

$$\Pr_{\boldsymbol{v}}\left[\phi_{\boldsymbol{v}}(\boldsymbol{F}_1) = \phi_{\boldsymbol{v}}(\boldsymbol{F}_2)\right] \le \frac{1}{|\mathbb{F}|},$$

and thus by Claim B.1 it holds that there exists a $\boldsymbol{v} \in \mathbb{F}^k$ such that

$$|S_{\boldsymbol{v}}| \ge \frac{|S|}{1 + (|S|-1)\cdot\frac{1}{|\mathbb{F}|}} = \frac{|\mathbb{F}|\cdot|\Lambda(\mathcal{C}^{\equiv_2}, \delta)|}{|\mathbb{F}| + |\Lambda(\mathcal{C}^{\equiv_2}, \delta)| - 1}.$$

Fixing this $\boldsymbol{v}$, for any $\mu_1, \mu_2 \in \mathbb{F}$, define the set Γ_{μ_1,μ_2} as follows:

$$\Gamma_{\mu_1,\mu_2} \coloneqq \{\gamma \in \mathbb{F} : \exists\, (a_1, a_2) \in S_{\boldsymbol{v}} \text{ s.t. } a_1 + \gamma\cdot a_2 = \mu_1 + \gamma\cdot\mu_2\}.$$

Note that, letting $\mathbb{x} = (\boldsymbol{v}, \mu_1, \mu_2)$ and $\mathbb{y} = (f_1, f_2)$, it holds that $\Gamma_{\mu_1,\mu_2} \subseteq \Omega_{\mathbb{x}}^{\mathbb{y}}$, where Ω is as in Definition 6.11. Further, if $(\mu_1, \mu_2) \notin S_{\boldsymbol{v}}$, it holds that $\Delta(\mathbb{y}, \mathcal{R}_{\mathcal{C}}^2[\mathbb{x}]) > \delta$, and so the soundness error of Construction 6.9 is lowerbound by

$$\frac{|\Gamma_{\mu_1,\mu_2}|}{|\mathbb{F}|} \text{ subject to } (\mu_1, \mu_2) \notin S_{\boldsymbol{v}}.$$

Let $S_{\boldsymbol{v}}^{\text{2nd}} = \{a_2 : (a_1, a_2) \in S_{\boldsymbol{v}}\}$. Note that $|S_{\boldsymbol{v}}^{\text{2nd}}| \le |S_{\boldsymbol{v}}| \le |\Lambda(\mathcal{C}^{\equiv_2}, \delta)| < |\mathbb{F}|$ by assumption, so there exists $\mu_2 \notin S_{\boldsymbol{v}}^{\text{2nd}}$. Fix such a μ_2 so that for any $\mu_1 \in \mathbb{F}$ we have that $(\mu_1, \mu_2) \notin S_{\boldsymbol{v}}$. Next, we choose μ_1 so that the map $\psi\colon S_{\boldsymbol{v}} \to \Gamma_{\mu_1,\mu_2}$ defined as:

$$\psi\colon S_{\boldsymbol{v}} \to \Gamma_{\mu_1,\mu_2}, \qquad (a_1, a_2) \mapsto \frac{\mu_1 - a_1}{a_2 - \mu_2},$$

is well-defined and injective. This suffices since it implies that $|\Gamma_{\mu_1,\mu_2}|/|\mathbb{F}| \ge |S_{\boldsymbol{v}}|/|\mathbb{F}| \ge \frac{|\Lambda(\mathcal{C}^{\equiv_2},\delta)|}{|\mathbb{F}|+|\Lambda(\mathcal{C}^{\equiv_2},\delta)|-1}$. First note that, since $\mu_2 \notin S_{\boldsymbol{v}}^{\text{2nd}}$, the denominator never vanishes, and for $(a_1, a_2) \in S_{\boldsymbol{v}}$ we have that

$$\gamma = \psi(a_1, a_2) = \frac{\mu_1 - a_1}{a_2 - \mu_2} \implies a_1 + \gamma\cdot a_2 = \mu_1 + \gamma\cdot\mu_2,$$

and thus $\psi(a_1, a_2) \in \Gamma_{\mu_1,\mu_2}$ as desired. We are left to show that the map is injective when μ_1 is chosen appropriately. Fix distinct pairs $(a_1, a_2), (b_1, b_2) \in S_{\boldsymbol{v}}$, and suppose that $\psi(a_1, a_2) = \psi(b_1, b_2)$, which implies that

$$\frac{\mu_1 - a_1}{a_2 - \mu_2} = \frac{\mu_1 - b_1}{b_2 - \mu_2}$$

Note that if $a_2 = b_2$ we have that $a_1 = b_1$, a contradiction to $(a_1, a_2) \neq (b_1, b_2)$, so we assume that $a_2 \neq b_2$. Then, by solving the above equation we obtain that μ_1 must be $\mu_1 = \frac{\mu_2\cdot(a_1-b_1)+a_1b_2-b_1a_2}{b_2-a_2}$. Define the set

$$\mathcal{B} \coloneqq \left\{ \frac{\mu_2\cdot(a_1 - b_1) + a_1b_2 - b_1a_2}{b_2 - a_2} \;\middle|\; (a_1, a_2), (b_1, b_2) \in \binom{S_{\boldsymbol{v}}}{2} \right\}.$$

Since $|\mathcal{B}| \le \binom{|S_{\boldsymbol{v}}|}{2} \le \binom{|\Lambda(\mathcal{C}^{\equiv_2},\delta)|}{2} < |\mathbb{F}|$ (where the last inequality follows by assumption), it must be that there exists $\mu_1 \in \mathbb{F}\setminus\mathcal{B}$, and then ψ is injective. $\square$

6.4.2 Correlated agreement lower bound

We show a large correlated agreement error directly leads to an attack on the soundness of Construction 6.9.

Lemma 6.13. *The soundness error of Construction 6.9 from $\tilde{\mathcal{R}}^2_{\mathcal{C},\delta}$ to $\tilde{\mathcal{R}}^1_{\mathcal{C},\delta}$ is at least $\varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$.*

Proof. Let $\delta \in (0,1)$. Suppose that $\varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) > 0$, if not the statement holds vacuously. Let f_1, f_2 be two functions maximizing the CA error, that is

$$\Pr_{\gamma \leftarrow \mathbb{F}}\left[\begin{array}{c} \Delta(f_1 + \gamma \cdot f_2, \mathcal{C}) \leq \delta \\ \wedge\, \Delta((f_1, f_2), \mathcal{C}^{\equiv 2}) > \delta \end{array}\right] = \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)\,.$$

Since $\varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) > 0$, it must be that $\Delta((f_1, f_2), \mathcal{C}^{\equiv 2}) > \delta$. Let $S = \{\gamma : \Delta(f_1 + \gamma \cdot f_2, \mathcal{C}) \leq \delta\}$, and note that $|S| = \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta) \cdot |\mathbb{F}|$. Clearly $S \subseteq \Omega^{f_1,f_2}_{\mathbf{0}^k,0,0}$, which concludes the lemma. □

Remark 6.14. We stress that Lemma 6.13 lower bounds the soundness of Construction 6.9 in terms of the correlated agreement error $\varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$, rather than the *mutual correlated agreement error* $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$ that appears in the soundness error in Lemma 6.10. A lower bound based on mutual correlated agreement would be qualitatively stronger, since $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta) \geq \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$. However, no attack that succeeds all the way to $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta)$, assuming $\varepsilon_{\mathsf{mca}}(\mathcal{C},\delta) > \varepsilon_{\mathsf{ca}}(\mathcal{C},\delta)$, is currently known. Note also that the attack in Lemma 6.13 *does not leverage* the triple $\mathbb{x} = (\boldsymbol{v}, \mu_1, \mu_2)$, indicating that exploiting clever choices of $\mathbb{x}$ might lead to sharper results (similar to Lemma 6.12).

7 Related problems and promising directions

In this section we briefly state open challenges and possible directions towards either resolving the grand challenges in the introduction or improving IOP and SNARK constructions.

Mutual correlated agreement for non-polynomial codes. This paper focuses on list-decoding and MCA bounds for Reed–Solomon codes and related polynomial codes, as those are the most typically deployed and studied codes. A long line of works considers constructions using instead *non-polynomial codes* with a focus on linear-time encodable codes (e.g., expander codes, turbo codes, sketched codes, and others) so as to enable linear-time provers (e.g., [BCGGHJ17; BCG20; BCL22; GLSTW23; BCFRRZ25; BMMS25a; BCFW25; BMMS25b; BFRW25]). For many of these codes, the best known list-decoding and MCA bounds are those proved for *general linear codes*, which are significantly weaker than what is known for Reed–Solomon codes. A natural line of research is to investigate whether better list-decoding and MCA bounds can be shown for these codes. This has implications that are both practical (leading to better succinct arguments) and theoretical (as many of these codes have *non-explicit* constructions, which suggests the techniques to prove improved list-decoding and MCA bounds might significantly differ from polynomial-codes). Some preliminary results in this direction can be found in [GG25], which showed that random LDPC codes and AEL codes have MCA up to *capacity.*

Characterization of degenerate codes. On a similar note, we know that some of the list-decoding and MCA bounds known for general linear codes are in fact *tight*: there exists codes for which no better list-decoding or MCA bound can be proven. Typically, these codes tend to be degenerate, and one could reasonably conjecture that any "well behaved code" achieves better bounds than the ones for generic linear codes. Understanding which codes are indeed well behaved and which ones are not is an open ended research question.

The effect of interleaving. The construction of an interleaved code $\mathcal{C}^{\equiv s}$ from a base code $\mathcal{C}$ has many applications to IOPs [AHIV17; GLSTW23; ACFY25; NA25; CFW26], and the soundness of the resulting construction depends on the list-decoding and MCA bounds of the *interleaved code* $\mathcal{C}^{\equiv s}$. Statements such as Lemma 2.10 and Lemma 4.7 show that these properties are intimately connected to the corresponding properties of the base code $\mathcal{C}$. The situation is not, however, completely satisfactory. In list-decoding, the list size of the interleaved code $\mathcal{C}^{\equiv s}$ is bounded from above by the list size of the base code $\mathcal{C}$ by a factor that is *independent* of the interleaving parameter s, but which grows exponentially when $\delta \to \delta_{\mathsf{min}}(\mathcal{C})$. The MCA error instead does not depend on δ, but yet grows *linearly* with the interleaving parameter s, i.e. $\varepsilon_{\mathsf{mca}}(\mathcal{C}^{\equiv s}, \delta) \leq s \cdot \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$. We do not know whether these bounds are tight in general; are there linear codes such that Lemma 2.10 and Lemma 4.7 are met with equality for all $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C}))$? An indication that the MCA bound is not tight comes in the work of [DG24], which shows that, when $\delta \in (0, \delta_{\mathsf{min}}(\mathcal{C})/2)$, we have that $\varepsilon_{\mathsf{mca}}(\mathcal{C}^{\equiv s}, \delta) = \varepsilon_{\mathsf{mca}}(\mathcal{C}, \delta)$. As a research direction, we propose both understanding whether the existing bounds are tight and characterizing whether for a chosen family of codes (such as the polynomial codes discussed in this paper or the non-polynomial codes mentioned earlier) tighter bounds can be derived.

Improved parameters for subspace-design codes. From the recent works of [CZ25; GG25], we know that subspace-design codes exhibit good list-decoding and MCA properties up to the capacity regime. The tradeoff is that for the subspace-design codes that we know of, to achieve good properties up to $\delta := \delta_{\mathsf{min}}(\mathcal{C}) - \eta$ the size of the alphabet must increase to be $s = O(1/\eta^2)$. Empirically (as investigated in Section 6.3) the increase in alphabet size eliminates the gain in argument size that larger proximity parameters typically translate to. To the best of our knowledge, no natural barriers to better dependencies between η and s are known, which might reverse the comparison in Section 6.3 and lead to concretely more efficient succinct arguments from subspace-design codes. Conversely, discovering such a barrier would indicate that the subspace-design route is unlikely to be beneficial for succinct arguments.

MCA for univariate powers. In this paper, we considered MCA for an affine combination of codewords, namely combinations of the form $f_0 + \gamma \cdot f_1$. The design of IOP often mandates more general types of combinations, such as *polynomial generators* (where the combination is generated by low-degree polynomials in the randomness sampled), and one can study MCA with respect to this class of combination. For example, one could study *curve MCA* for univariate combinations of the form $\sum_{i=0}^{\ell} \gamma^i \cdot f_i$. The work of [BCGM25] shows that a linear code $\mathcal{C}$ that has MCA with respect to both affine combinations and univariate combinations is sufficient to show that the $\mathcal{C}$ has good MCA properties with respect to *every* polynomial generator. Understanding which codes have these two properties is an open question.

Derandomizing Reed–Solomon results. An important piece of evidence that Reed–Solomon codes may have good list-decoding and MCA properties up to capacity is the fact that *randomly punctured* Reed–Solomon codes exhibit such properties (Theorem 4.15). One possible direction to show that explicit Reed–Solomon codes have the same properties is to *derandomize* such results. More precisely, these results show that (for any rate ρ, a sufficiently large field $\mathbb{F}$, and message length k), the code $\mathcal{C} \coloneqq \mathsf{RS}[\mathbb{F}, \mathcal{L}, k]$, where $\mathcal{L} \leftarrow \binom{\mathbb{F}}{k/\rho}$, has good list-decoding and MCA properties. Instead of choosing the subset $\mathcal{L}$ uniformly at random in the field $\mathbb{F}$, one can reasonably consider the case where the subset $\mathcal{L}$ has more structure, and whether similar properties hold for the resulting codes.

A Additional preliminaries

In this section we introduce further concepts that are used throughout this paper.

A.1 Interactive oracle reductions

Relations. In this paper a relation is a set of triples $(\mathbb{x}, \mathbb{y}, \mathbb{w})$:

- $\mathbb{x}$ is the *explicit instance* of the relation.
- $\mathbb{y}$ is the *implicit instance* of the relation.
- $\mathbb{w}$ is the *witness* of the relation.

Interactive Oracle Reductions (IORs) [BCGGRS19; BMNW25] are a generalization of Interactive Oracle Proofs (IOPs) [BCS16; RRR16]. Below we describe the flavor of IOR relevant to this paper.

A public-coin IOR of proximity from relation $\mathcal{R}$ to relation $\mathcal{R}'$ is a tuple $\mathsf{IOR} = (\mathbf{P}, \mathbf{V})$ that works as follows. The prover $\mathbf{P}$ is an interactive algorithm that receives as input $(\mathbb{x}, \mathbb{y}, \mathbb{w}) \in \mathcal{R}$, and the verifier $\mathbf{V}$ is an interactive oracle algorithm that receives input $\mathbb{x}$ and query access to $\mathbb{y}$ (consisting of l_0 symbols over some alphabet Σ_0). They interact over k rounds: in each round $i \in [\mathsf{k}]$, $\mathbf{P}$ sends a proof string π_i consisting of l_i symbols over some alphabet Σ_i, and $\mathbf{V}$ sends a uniformly random message $\rho_i \in \{0,1\}^{\mathsf{r}_i}$. After k rounds of interaction, $\mathbf{V}$ makes queries to $\mathbb{y}, \pi_1, \ldots, \pi_{\mathsf{k}}$ (each query to a symbol). After that: (i) $\mathbf{V}$ either rejects or outputs a new explicit instance $\mathbb{x}'$ and implicit instance $\mathbb{y}'$ (which is implicitly specified in terms of the oracles $\mathbb{y}$ and $\pi_1, \ldots, \pi_{\mathsf{k}}$), and (ii) $\mathbf{P}$ outputs a new witness $\mathbb{w}'$ such that $(\mathbb{x}', \mathbb{y}', \mathbb{w}') \in \mathcal{R}'$. We write $\langle \mathbf{P}, \mathbf{V} \rangle(\mathbb{x}, \mathbb{y}, \mathbb{w})$ for the interaction of $\mathbf{P}, \mathbf{V}$ above.

Definition A.1. $\mathsf{IOR} = (\mathbf{P}, \mathbf{V})$ *has* **(perfect) completeness** *if, for every* $(\mathbb{x}, \mathbb{y}, \mathbb{w}) \in \mathcal{R}$,

$$\Pr\left[(\mathbb{x}', \mathbb{y}', \mathbb{w}') \in \mathcal{R}' \;\middle|\; (\mathbb{x}', \mathbb{y}', \mathbb{w}') \leftarrow \langle \mathbf{P}, \mathbf{V} \rangle(\mathbb{x}, \mathbb{y}, \mathbb{w}) \right] = 1 \,.$$

Remark A.2. An IOP for a relation $\mathcal{R}$ is an IOR from the relation $\mathcal{R}$ to the trivial relation $\mathcal{R}_\perp := \{(\mathbb{x} = \perp, \mathbb{y} = \perp, \mathbb{w} = \perp)\}$.

Knowledge soundness for IORs. Applications of IORs typically require a strengthening of the classical notion of soundness, known as knowledge soundness.

Definition A.3. *An interactive oracle reduction* IOR $= (\mathbf{P}, \mathbf{V})$ *has* **knowledge soundness from $\tilde{\mathcal{R}}$ to $\tilde{\mathcal{R}}'$ with error κ and extraction time** et *if there exists an extractor* $\mathbf{E}$ *that runs in time* et *such that for every statement* $(\mathbb{x}, \mathbb{y})$*, and unbounded malicious prover* $\tilde{\mathbf{P}}$*,*

$$\Pr_{\rho_1, \ldots, \rho_{\mathsf{k}}}\left[\begin{array}{c} (\mathbb{x}', \mathbb{y}', \mathbb{w}') \in \tilde{\mathcal{R}}' \\ \wedge\; (\mathbb{x}, \mathbb{y}, \mathbb{w}) \notin \tilde{\mathcal{R}} \end{array} \;\middle|\; \begin{array}{l} (\mathbb{x}', \mathbb{y}', \mathbb{w}') \xleftarrow{\mathsf{tr}} \langle \tilde{\mathbf{P}}, \mathbf{V} \rangle(\mathbb{x}, \mathbb{y}) \\ \mathbb{w} \leftarrow \mathbf{E}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathbb{x}', \mathbb{y}', \mathbb{w}') \end{array} \right] \leq \kappa(\mathbb{x}, \mathbb{y}) \,.$$

Sometimes we leave implicit some parameters of κ if they are clear from context.

The IORs that we consider in this paper satisfy a strong notion of knowledge soundness, namely round-by-round knowledge soundness [BCFW25], that guarantees security after the [FS86] transformation.

Definition A.4. *Let* $\mathsf{IOR} = (\mathbf{P}, \mathbf{V})$ *be an IOR from a relation* $\mathcal{R}$ *to* $\mathcal{R}'$. *A* **knowledge state function from** $\tilde{\mathcal{R}}$ **to** $\tilde{\mathcal{R}}'$ *for* IOR *is a (possibly inefficient) function* State *that, on input a statement* $(\mathbb{x}, \mathbb{y})$, *interaction transcript* tr, *and* **knowledge state witness** w, *outputs a bit and has the following properties.*

- *Empty transcript: if* $\mathsf{tr} = \emptyset$ *is the empty transcript, then* $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 1$ *if and only if* $(\mathbb{x}, \mathbb{y}, \mathsf{w}) \in \tilde{\mathcal{R}}$.
- *Prover moves: if* tr *is a transcript where the prover is about to move and* $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 0$, *then* $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\pi, \mathsf{w}) = 0$ *for every prover message* π.
- *Full transcript: if* $\mathsf{tr} = (\pi_1, \rho_1, \ldots, \pi_{\mathsf{k}}, \rho_{\mathsf{k}})$ *is a full transcript, then* $\mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathsf{w}) = 1$ *if and only if* $\mathbf{V}^{\mathbb{y}, \pi_1, \ldots, \pi_{\mathsf{k}}}(\mathbb{x}, \rho_1, \ldots, \rho_{\mathsf{k}})$ *outputs* $(\mathbb{x}', \mathbb{y}')$ *such that* $(\mathbb{x}', \mathbb{y}', \mathsf{w}) \in \tilde{\mathcal{R}}'$.

Definition A.5. $\mathsf{IOR} = (\mathbf{P}, \mathbf{V})$ *from a relation* $\mathcal{R}$ *to* $\mathcal{R}'$ *has* **round-by-round knowledge soundness from** $\tilde{\mathcal{R}}$ **to** $\tilde{\mathcal{R}}'$**, errors** $(\varepsilon_1, \ldots, \varepsilon_{\mathsf{k}})$**, and extraction times** $(\mathsf{et}_1, \ldots, \mathsf{et}_{\mathsf{k}})$ *if there exist a knowledge state function* State *with relaxation* $\tilde{\mathcal{R}}$ *to* $\tilde{\mathcal{R}}'$ *for* IOR *and a deterministic extractor* $\mathbf{E}_{\mathrm{rbr}}$ *with the following property: for every statement* $(\mathbb{x}, \mathbb{y})$, *round index* $i \in [\mathsf{k}]$, *and interaction transcript* $\mathsf{tr} = (\pi_1, \rho_1, \ldots, \pi_{i-1}, \rho_{i-1}, \pi_i)$ *where the verifier is about to move,* $\mathbf{E}_{\mathrm{rbr}}$ *runs in time at most* et_i *and*

$$\Pr_{\rho_i}\left[\exists \mathsf{w} : \begin{array}{l} \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}, \mathbf{E}_{\mathrm{rbr}}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\rho_i, \mathsf{w})) = 0 \\ \wedge\ \mathsf{State}(\mathbb{x}, \mathbb{y}, \mathsf{tr}\|\rho_i, \mathsf{w}) = 1 \end{array}\right] \leq \varepsilon_i(\mathbb{x}, \mathbb{y}) \,.$$

The **total extraction time** *is* $\sum_{i \in [\mathsf{k}]} \mathsf{et}_i$.

A.2 Univariate multiplicity codes

Univariate multiplicity codes instead choose each alphabet symbol $\mathbb{F}^s$ to bundle the evaluations of the s *formal derivatives* of the message. We recall the formal derivative of a polynomial.

Definition A.6. *Let* $k \in \mathbb{N}$, $\mathbb{F}$ *be a finite field with* $\mathsf{char}(\mathbb{F}) \geq k$. *For a polynomial* $\hat{f} \in \mathbb{F}^{<k}[X]$ *with* $\hat{f}(X) = \sum_{i=0}^{k-1} a_i \cdot X^i$ *we define the* **formal derivative polynomial** $\hat{f}' \in \mathbb{F}^{<k-1}[X]$ *as*

$$\hat{f}'(X) = \sum_{i=1}^{k-1} (a_i \cdot i) \cdot X^{i-1} \,.$$

For $s \in \mathbb{N}$ *we also define recursively* $\hat{f}^{(0)} = \hat{f}$ *and* $\hat{f}^{(s)} = (\hat{f}^{(s-1)})'$.

Using the formal derivative, we can define multiplicity codes. Note that, compared to folded Reed–Solomon codes, univariate multiplicity codes require the characteristic of the field to be larger than the message size (but have looser requirements on the size of the field itself).

Definition A.7 ([GW13; KSY14])**.** *Let* $n, k, s \in \mathbb{N}$, $\mathbb{F}$ *be a finite field with* $\mathsf{char}(\mathbb{F}) \geq k$ *and* $|\mathbb{F}| \geq n$ *and let* $\mathcal{L} \subseteq \mathbb{F}$ *with* $|\mathcal{L}| = n$. *We define the* **univariate multiplicity code***:*

$$\mathsf{UM}[\mathbb{F}, \mathcal{L}, k, s] \coloneqq \left\{ f \colon \mathcal{L} \to \mathbb{F}^s \;\middle|\; \exists \hat{f} \in \mathbb{F}^{<k}[X],\ \forall x \in \mathcal{L} \colon f(x) = \begin{bmatrix} \hat{f}^{(0)}(x) \\ \hat{f}^{(1)}(x) \\ \vdots \\ \hat{f}^{(s-1)}(x) \end{bmatrix} \right\} .$$

Note that $\mathsf{UM}[\mathbb{F},\mathcal{L},k,1]=\mathsf{RS}[\mathbb{F},\mathcal{L},k]$ and that further for any message $\hat{f}\in\mathbb{F}^{<k}[X]$ we have that $\mathsf{UM}[\mathbb{F},\mathcal{L},k,s](\hat{f})=\mathsf{IRS}[\mathbb{F},\mathcal{L},k,s](\hat{f}^{(0)},\ldots,\hat{f}^{(s-1)})$, so the encoding can be performed with (roughly) the same concrete efficiency as with an interleaved Reed–Solomon code.

B Omitted claim for Lemma 6.12

Claim B.1. *Let S,T be finite sets and let Φ be a distribution on functions from S to T such that for any distinct $x,y\in S$,*

$$\Pr_{\phi\leftarrow\Phi}[\phi(x)=\phi(y)]\le\epsilon\,.$$

Then there exists some ϕ in the support of Φ such that

$$|\phi(S)|\ge\frac{|S|}{1+(|S|-1)\cdot\epsilon}\,.$$

Proof. Let $C_\phi=\left\{(x,y)\in\binom{S}{2}\ :\ \phi(x)=\phi(y)\right\}$ be the set of (distinct) colliding elements of S under ϕ. Note in particular that $\mathbb{E}_{\phi\leftarrow\Phi}[|C_\phi|]\le\binom{|S|}{2}\cdot\epsilon$ since

$$\mathbb{E}_{\phi\leftarrow\Phi}[|C_\phi|]=\sum_{(x,y)\in\binom{S}{2}}\Pr_{\phi\leftarrow\Phi}[\phi(x)=\phi(y)]\le\binom{|S|}{2}\cdot\epsilon\,.$$

Clearly, $|S|=\sum_{\mu\in\phi(S)}|\phi^{-1}(\mu)|$, and further (since each $\mu\in\phi(S)$ contributes $\binom{|\phi^{-1}(\mu)|}{2}$ colliding pairs) it must be that

$$|C_\phi|=\sum_{\mu\in\phi(S)}\binom{|\phi^{-1}(\mu)|}{2}=\frac{1}{2}\sum_{\mu\in\phi(S)}(|\phi^{-1}(\mu)|-1)\cdot|\phi^{-1}(\mu)|=\frac{1}{2}\left(-|S|+\sum_{\mu\in\phi(S)}|\phi^{-1}(\mu)|^2\right)\,.$$

Rearranging, multiplying through by $|\phi(S)|$ (and applying Cauchy–Schwarz), we obtain that

$$|\phi(S)|\cdot(2|C_\phi|+|S|)=|\phi(S)|\cdot\sum_{\mu\in\phi(S)}|\phi^{-1}(\mu)|^2\ge\left(\sum_{\mu\in\phi(S)}|\phi^{-1}(\mu)|\right)^2=|S|^2\,,$$

and thus $|\phi(S)|\ge\frac{|S|^2}{2|C_\phi|+|S|}$. By taking expectations, and by Jensen's we get that

$$\mathbb{E}_\Phi[|\Phi(S)|]\ge\mathbb{E}_\Phi\left[\frac{|S|^2}{2|C_\Phi|+|S|}\right]\ge\frac{|S|^2}{2\cdot\mathbb{E}_\Phi\left[|C_\Phi|\right]+|S|}\ge\frac{|S|^2}{2\cdot\binom{|S|}{2}\cdot\epsilon+|S|}\,.$$

Rearranging, we get the final bound, and this guarantees the existence of a ϕ with the desired properties. □

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